ACO – World Leaders Drainage Technology








NIHVA TECHNOLOGIES PVT LTD – SALES TRAINING

Dated : 13TH JUNE 2015

ACO – Worldwide Locations









ACO locations – at home on four continents

The backbone of our global presence consists of our 29 production sites in places like Germany, United Kingdom, Czech Republic, Russia, USA and China. ACO also operates sales & marketing offices in more than 40 countries. This helps us ensure that we always see the markets closely, through the eyes of our customers, and can supply them with the optimal solutions for the projects.

ACO Group - Overview

ACO Group facts and figures

- ACO was founded in 1946 and is headquartered in Rendsburg / Büdelsdorf, Germany
- ACO Group is a world market leader in drainage technology.
- With its integrated approach, ACO stands for professional drainage, efficient cleaning, and the controlled discharge or reuse of water.
- Products include drainage channels and drains, oil and grease separators, backflow stop systems, pumps and pressure-water-tight cellar windows and light shafts.
- The major innovation strength of the ACO Group is built on intense research and development, and its technical expertise in processing polymer concrete, plastic, cast iron, stainless steel and reinforced concrete.



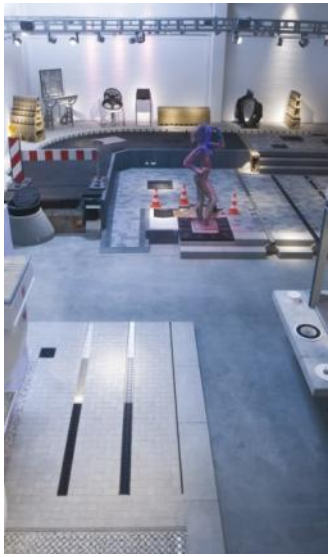


ACO Group at a glance

- 1946, Company founded by Josef-Severin Ahlmann
- 3,900 employees in more than 40 countries.
- 1,500 employees work at locations in Germany
- 29 production sites in 15 countries.
- Sales 2013: Euro 650 million.

ACO philosophy: Sharing knowledge means spreading knowledge

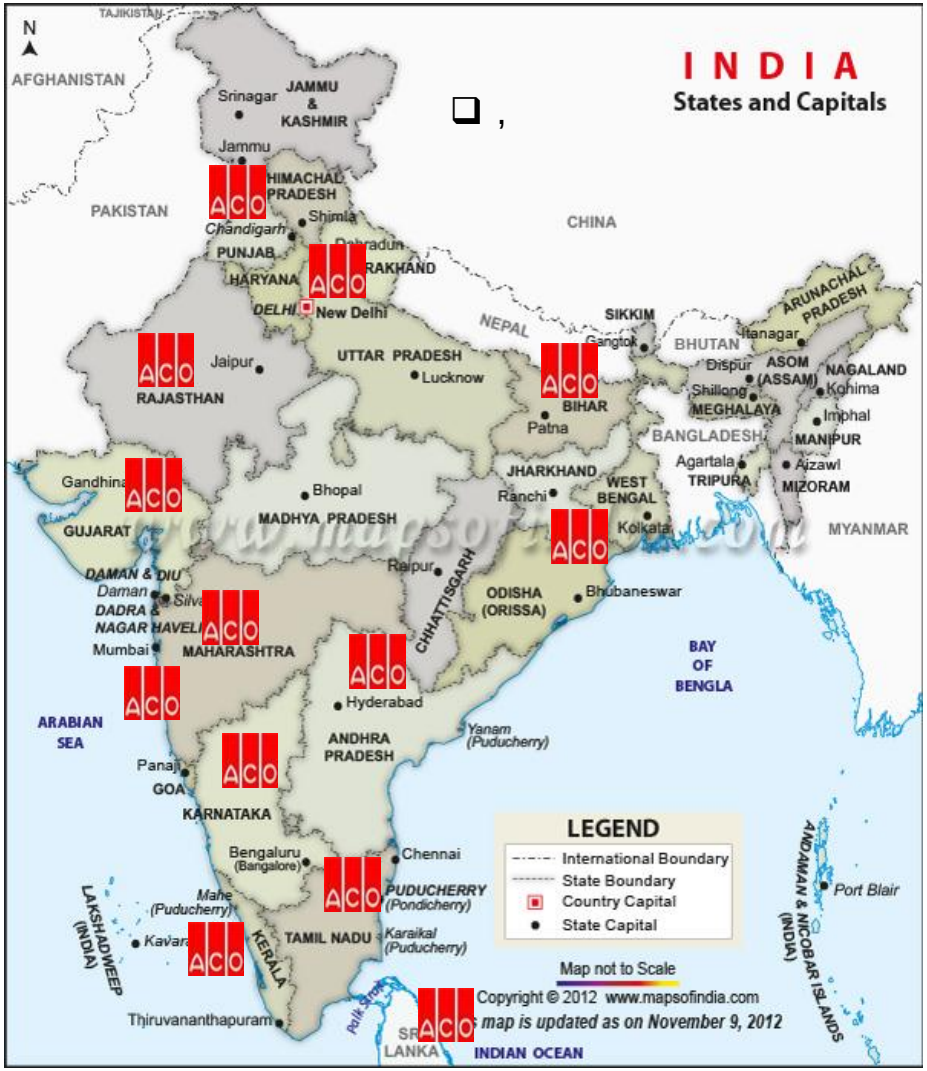



ACO Academy

- The ACO Academy in Rendsburg / Büdelsdorf (Germany) is our center for exchange of ideas and expertise amongst ACO employees, partners and the clients.
- The center houses Product displays, seminar halls, e-learning in virtual classrooms and of course facilitates business meetings in the Academy Lounge.

ACO INDIA- Pan India Network



- ACO Systems and Solutions Pvt. Ltd. is a 100% subsidiary of ACO Severin Ahlmann GmbH & Co. KG. Germany.
- ACO Systems and Solutions was incorporated in the year 2006 and is headquartered in Bangalore
- A team of dedicated and trained marketing and technical professionals located across the country to provide a complete range of Products and Solutions to various market verticals
- Equipped with partners across the country.
- Equipped with following service capabilities
  - End to End Engineering solutions
  - Advise on appropriate products and solutions
  - Pre and Post installation services
  - Installation supervision

ACO System Chain





ACO Products and Solutions

Classification by Product Group

EXTERNAL DRAINAGE

- Surface Water Drainage Systems
- Storm Water Management
- Sports Special
- Tunnel Drainage Systems
- Stormbrixx

BUILDING DRAINAGE

- Roof Drainage
- Balcony Drainage
- Façade Drainage
- Bath Drainage
- Kitchen Drainage
- Piping Systems

LANDSCAPE SOLUTIONS

- Kerb Drains
- Grass Grids
- Tree Guards
- Light Channel
- Brickslot
- Access Covers

SEPERATORS

- Kitchen Grease Seperators
- Petrol and Light Mineral Oil Seperators

EXTERNAL DRAINAGE SYSTEM

Current Solutions to 'Collect'











Effects of current solutions









Current Indian Scenario







Why Pre-fabricated Drains?



Why Pre-fabricated Drains?





External Drainage

Polymer Concrete Channel Drainage System



Application Areas

- Ramp Drainage
- Entrances
- Road Crossings
- Basement Drainage
- Car Parking

Polymer Concrete is a mixture of Polymers, Resins, Adhesives , Silicon and Mineral aggregates.



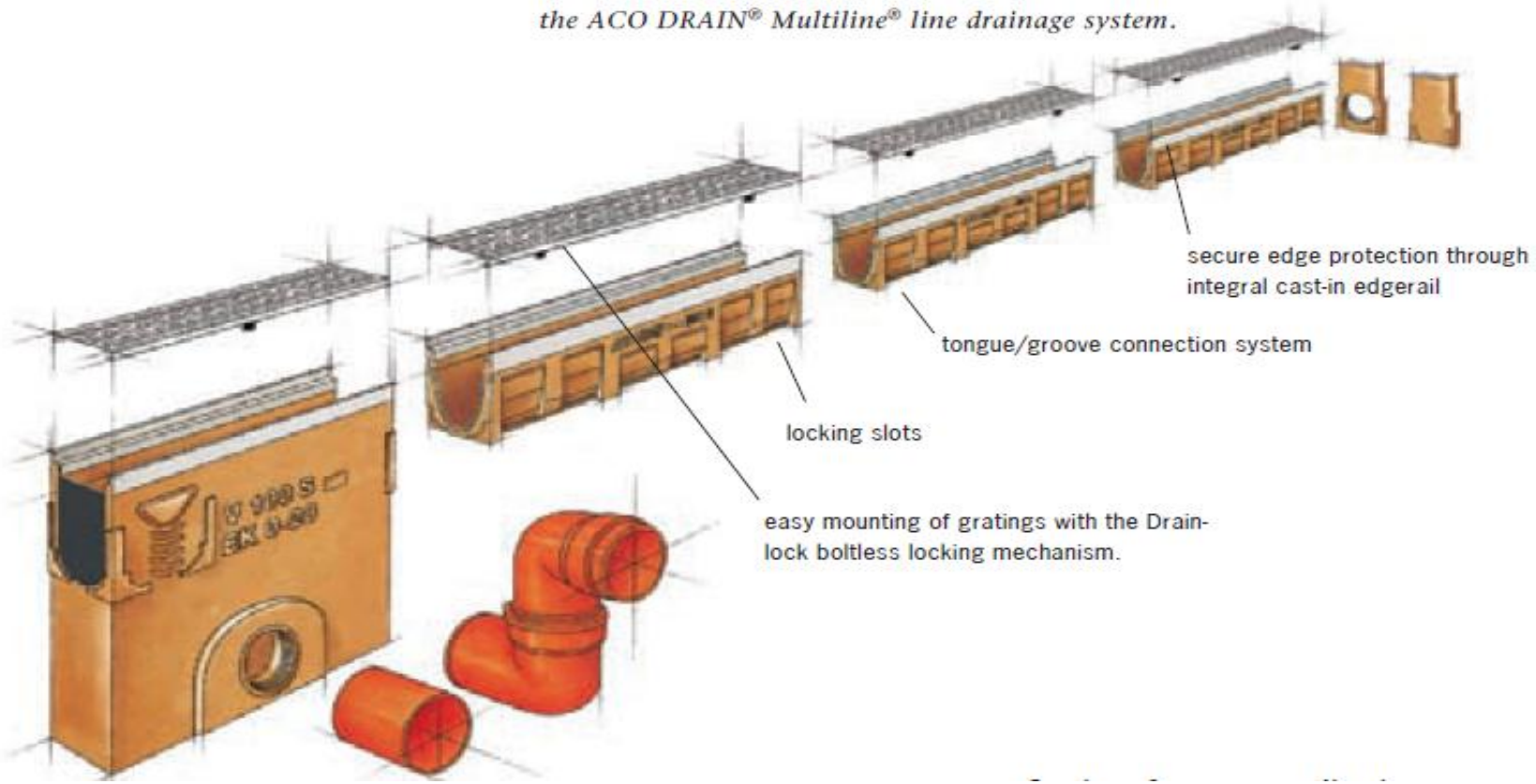




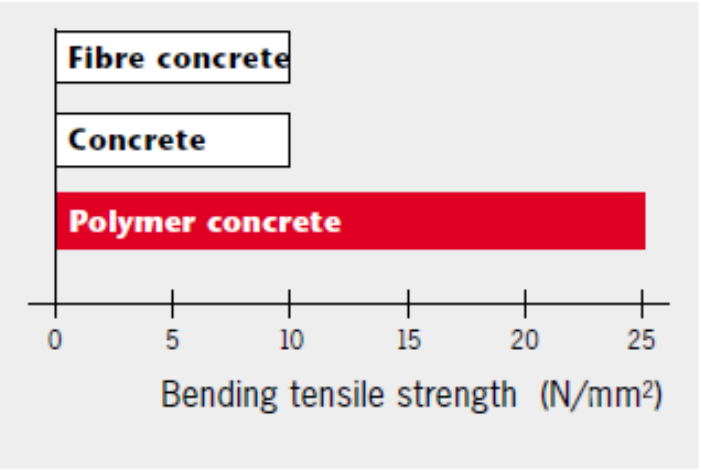


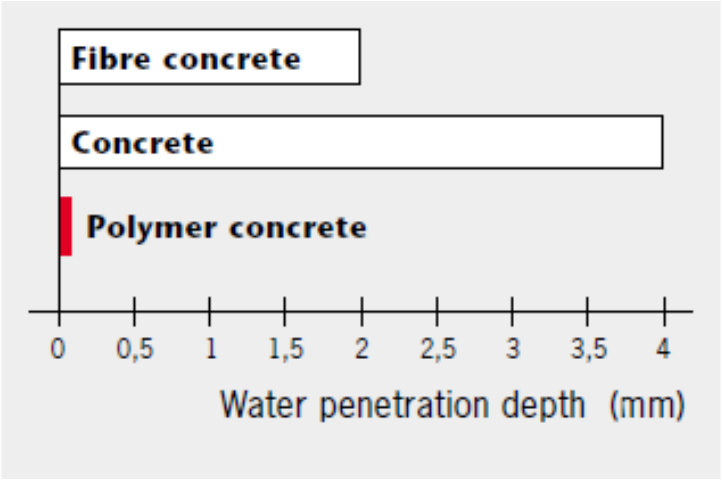
Freedom Of Design through Product Versatility

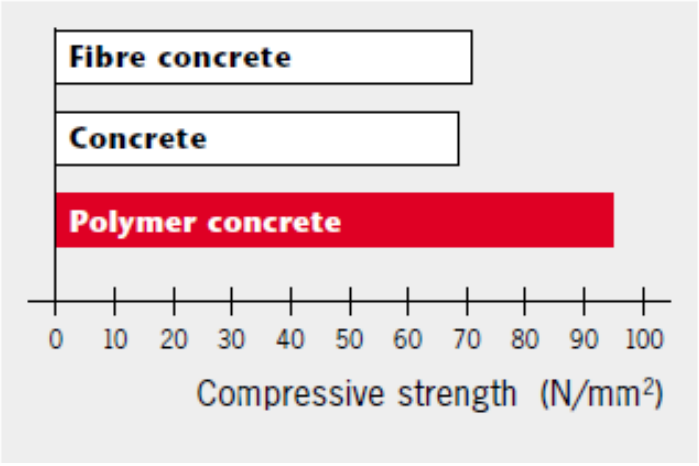
The ACO DRAIN® programme is a genuine modular system: individual, personalised solutions can be combined from a range of channels, gratings and system accessories such as sump boxes – a system which will convince you in terms of technology and economy. Take for example the ACO DRAIN® Multiline® line drainage system.

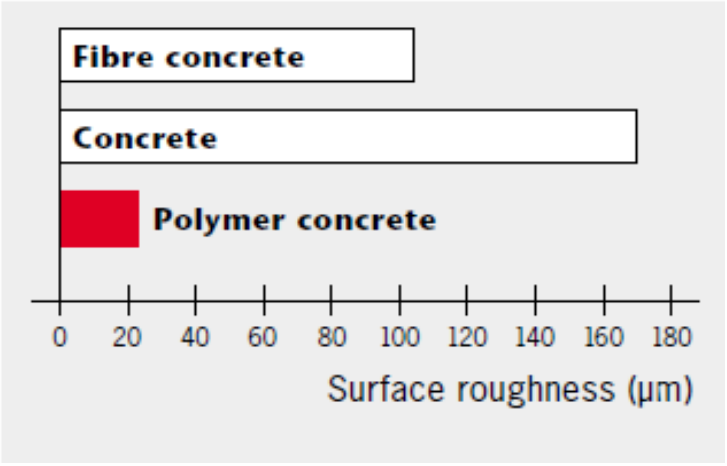


Why Polymer Concrete As Base Material For Drains?









- Light weight
- Age resistant
- Stable
- Rust free
- Frost, de-icing salt, chemical resistant

Salient Features of ACO Polymer Concrete Channels

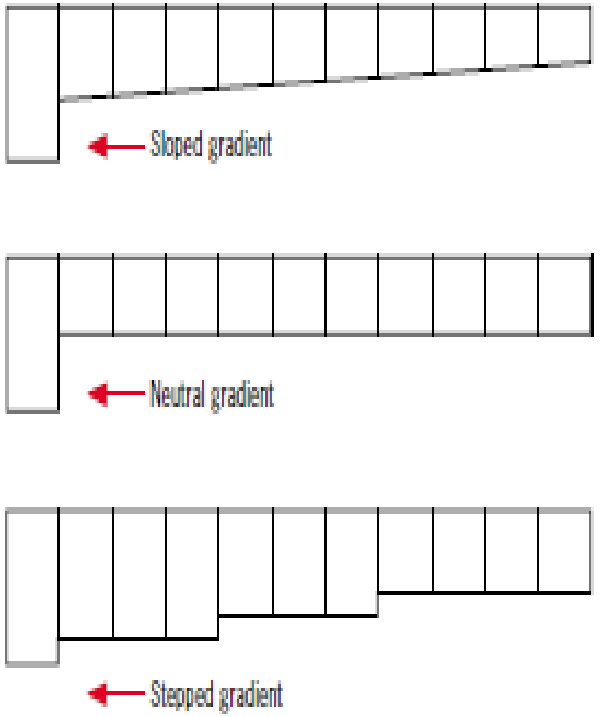


V Profile for
Hydraulic Efficiency



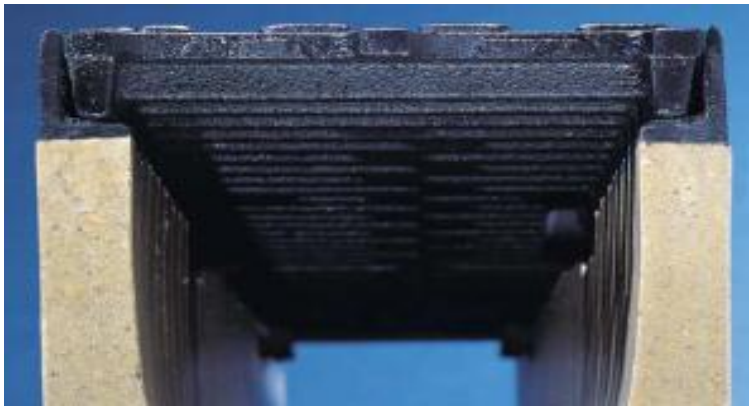


100 % Water Tight



Sloped Channels for
Enhanced Water Flow

Edge Rail Polymer Concrete & Load Class Lockable Gratings



Integrated edge-rails provide reliable edge protection.

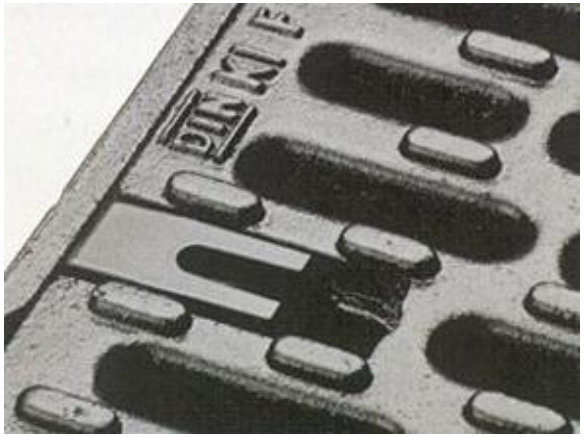






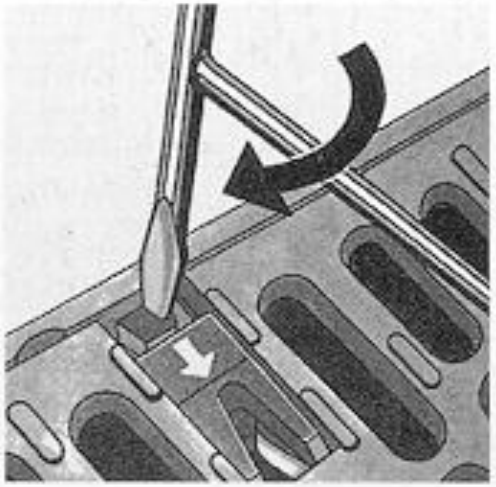
- Load class A 15
- Load class B 125
- Load class C 250
- Load class D 400
- Load class E 600
- Load class F 900

Power Drain – PowerLok™ locking system

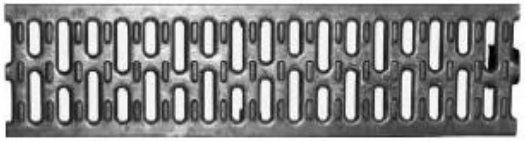


- **PowerLok™ : ACO Drain's patented, boltless locking system**
  - Provides quick fitting & removal of grates
  - Reduces installation & maintenance time and cost
  - No Rattling Noise on Gratings
- **Ductile Iron Edge Rail: Integrally cast ductile iron edge rail**

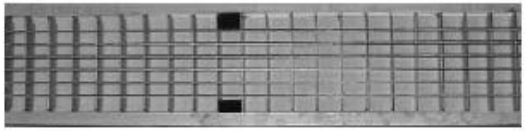




Grating Options



Bar grating



Mesh grating



522DL
Heelguard™ composite



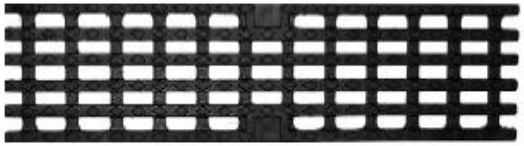
Perforated grating



Cast iron cover plate



423DL / 424DL
Slotted galvanised steel



Longitudinal rod grating

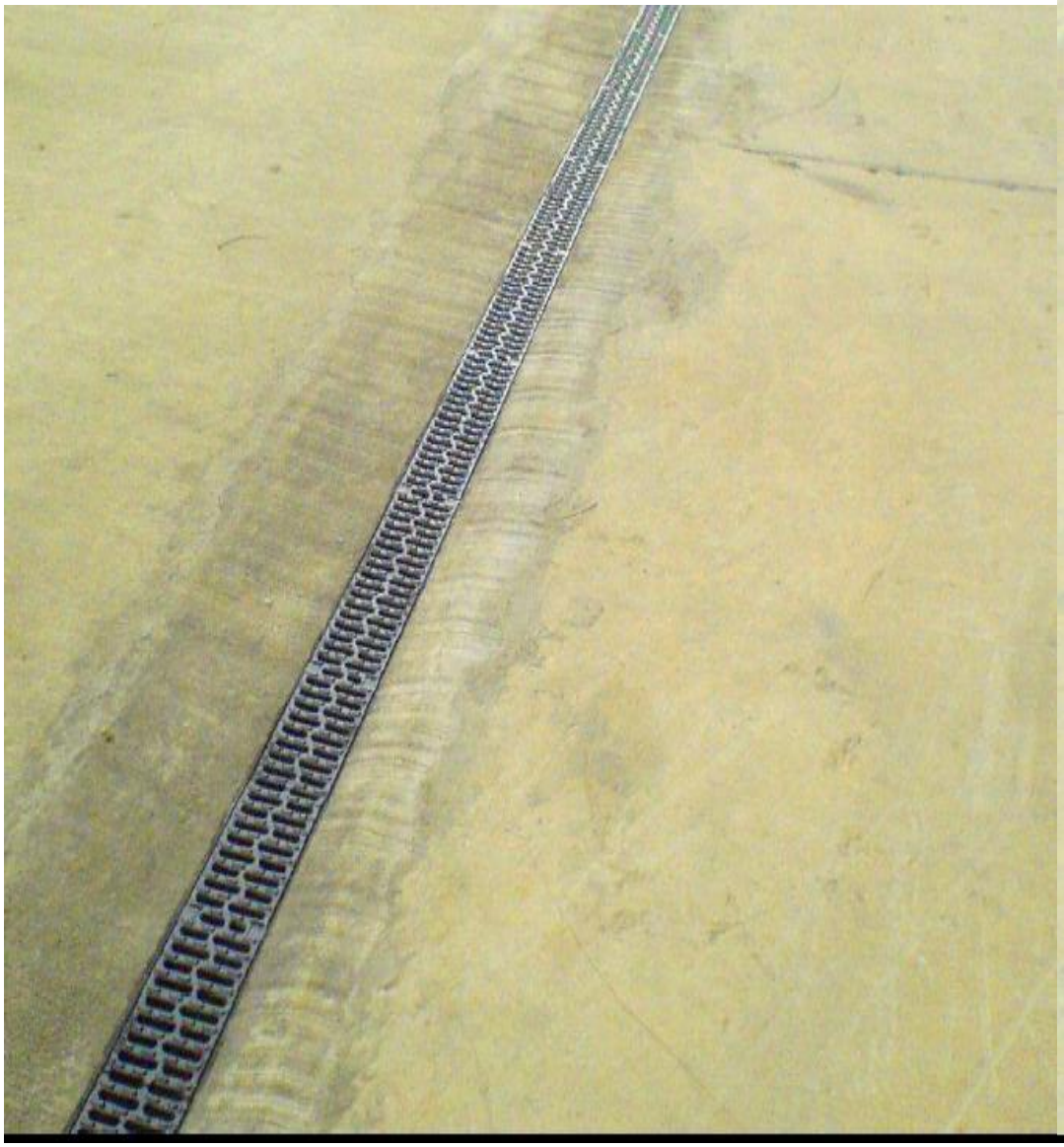
A Comparison

s.n.	ACO Channels	Conventional Channel
1	Gratings are Lockable. So more safer. Prevents accidents	Gratings are not Lockable. Hence vandalism can happen. If any grate is stolen accidents can happen
2	No Rattling of gratings on vehicular moment.	Rattling of Gratings on Vehicular moment.
3	No rattling hence no damage to grating as well as channels. Hence it is maintenance free.	Rattling will damage the grating as well as channels. Hence maintenance is required for the channels.
4	Better Hydraulics due to V profile	The hydraulics is poor.
5	Quick & easy instalaltion. Labour saving approx. 60%	The instalaltion is time consuming
6	Less depth so less excavation is needed. Almost 1/3rd of conventional drain	More excavation
7	Slope inbulit in the channel itself.	The channel has to be constructed in slope continuously
8	More sleeker channel will enhance the aesthetics of the instalation.	
9	Life expentancy of the product is almost 50 years if instalaled s per manufacturer instructions.	
10	The polymer concrete channel can be made 100% watertight by using polysulphide based sealent.	
11	No additional waterproofing is required for polyment concrete channel.	
12	100% green conpliance.	

External Drainage – Acropolis Project



External Drainage – Infosys





External Drainage – Infosys Bangalore





External Drainage – Infosys Bangalore

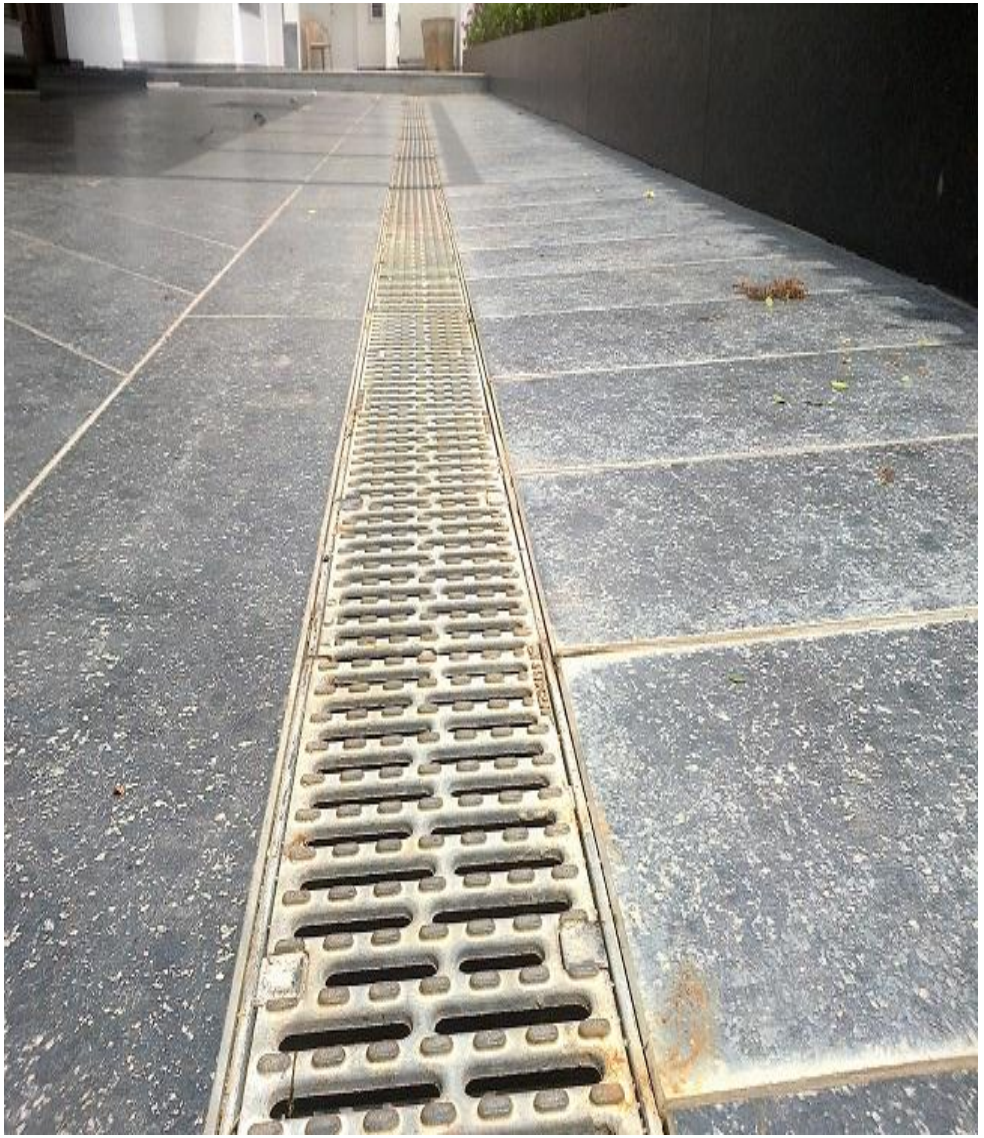


External Drainage



External Drainage



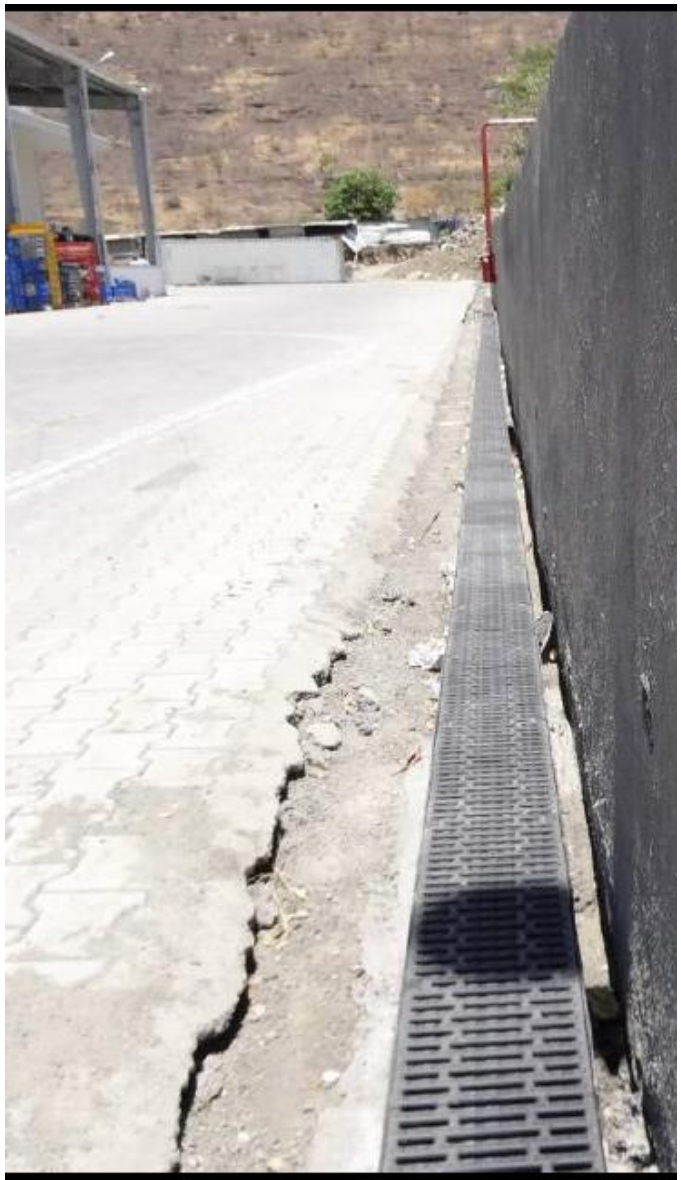


External Drainage – Raheja Group





External Drainage – Knorr Bremse Hinjewadi





External Drainage – P.R.Builders

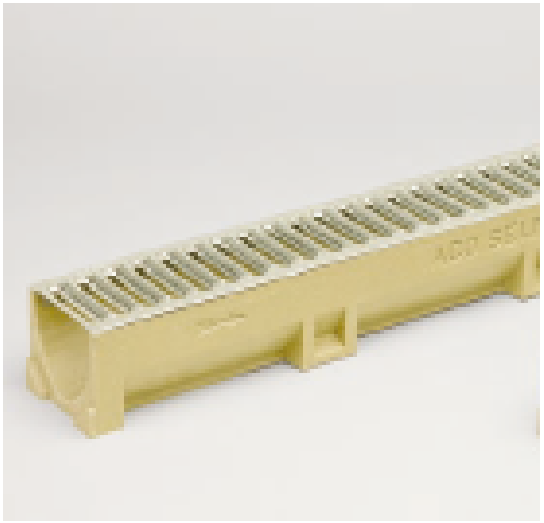




Podium Drainage / Terrace Drain

Surface Water Drainage Systems – Euroline / Supermini

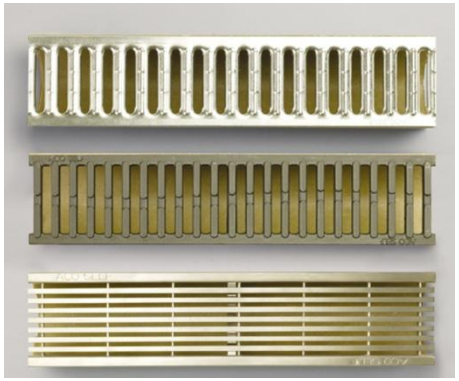
Pedestrain Load Class – For Balcony , Podium ,Terrace











- **ACO Self® EuroLine channel system finds excellent applications for use in Podium Drains for 100% Water Tight Application.**
- **The optimised channel body and the new locking system increase the channel's durability and safety, and make it easier to install.**
- **Total Installation Height from 65MM to 110MM**

Euroline – Gratings Options

Slot grating
Galvanized steel



Slot grating powder coated
Color: anthracite



Slot grating
Stainless steel electro polished



Slot grating powder coated
Color: Terracotta



Cast iron grating



Plastic grating with antislip
Microgrip



Brickslot
Stainless steel and galvanized steel



Wedge wire grating
Stainless steel



Podium Drain Installation – Cama Vandan Balconies





Podium Drain Installation – Cama Vandan Balconies





Podium /Terrace Drain Installation – Infosys

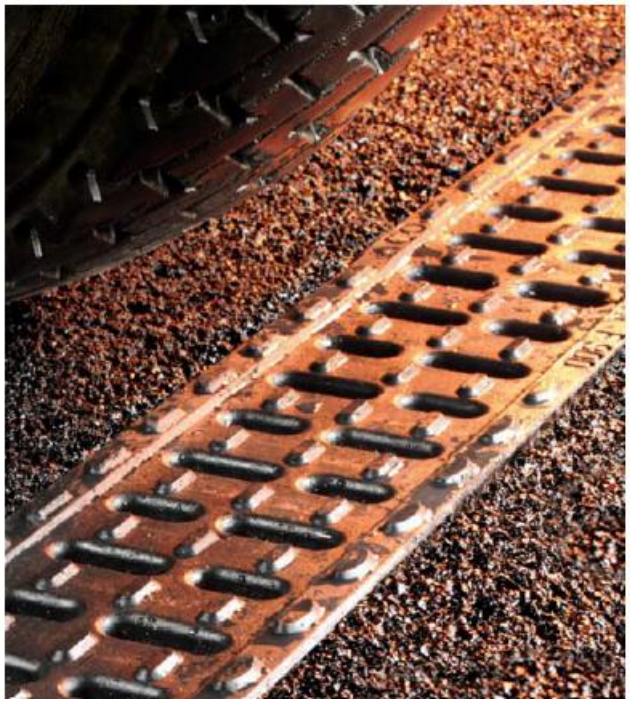






Surface Water Drainage Systems – ACO Road Drainage

Load Class – D 400 / E600 / F900







External Drainage – Q max Series

Storm Water Drainage Systems – High Capacity Drains

Heavy Load Class – For Draining Storm Water, Higher Volume of Drainage













Q MAX - Storm Water Drainage Systems







The ACO Qmax line drainage system was developed to satisfy demands for economical high capacity
drainage systems for large catchment areas.

ACO Qmax has passed independent load tests to
class F 900 in accordance with EN 1433.

Storm Water Drainage Systems

Manufactured from tough, chemically resistant medium density polyethylene (MDPE), ACO Qmax is light, easy to handle and quick to install



High hydraulic volume ACO Q*max* system in combined drainage and storage operation provides SuDS (Sustainable Drainage Systems) optional attenuated source control of storm water to prevent downstream flooding



Dry conditions, no rainfall and no outflow.



Rainfall starts, channel begins to fill with water and output via the flow regulator.



Water level in Qmax® system rises as storm intensity increases. Outflow reaches designed regulated flow rate without flooding of surrounding area.



Storm passes, controlled outflow continues and water level in channel system falls.

External Drainage – Aeroflex Project







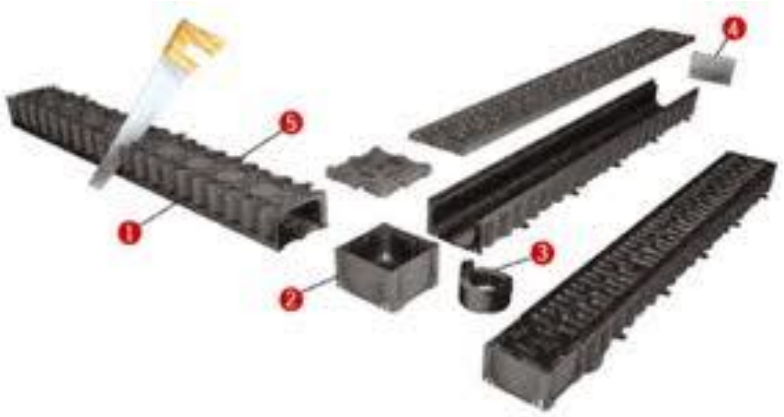
External Drainage – Polyproplene Drain

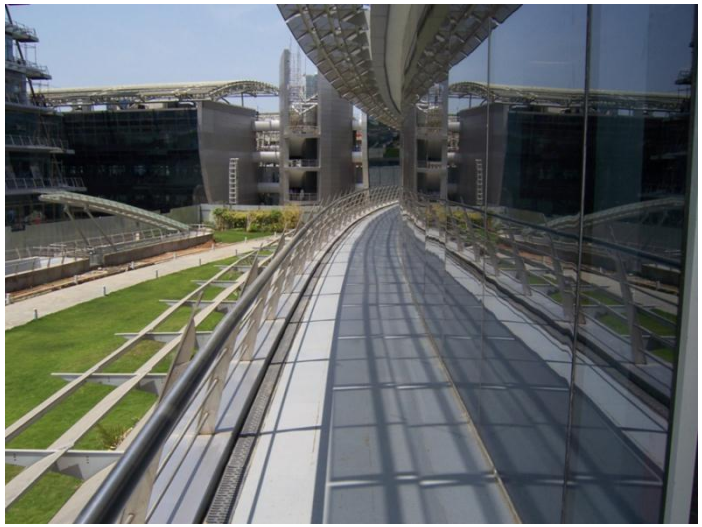
Surface Water Drainage Systems – Polypropylene Drains

Low Load Class – For Walkways, Parking , Club House etc















Polypropylene Channels

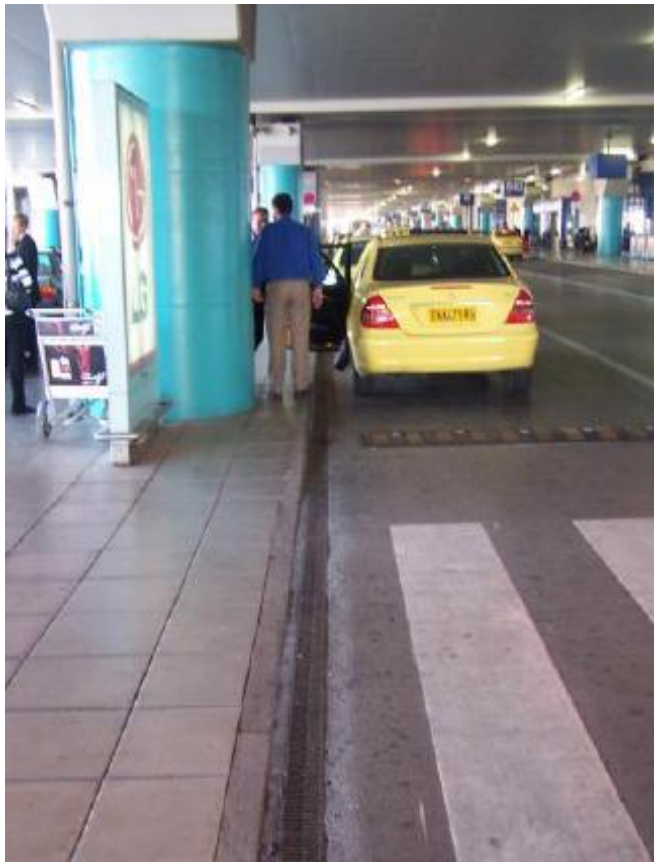


Specification

- Made of Poly Propylene
- No Slope Channels
- Load class from Pedestrian to Light duty vehicles
- Size 100 mm ,150 & 200 mm

Application Areas

- Peripheral Drainage
- Facades
- Basement Drainage



ACO Polypropylene Channels













- Available in 100,150 and 200mm width
- Wide range of Grating options to choose fro

Road Drainage









Road Drainage – Multi Top



Multitop® gully gratings for road drains

- For Load class D 400 Applications
- Sizes 300 x 500 mm and 500 x 500 mm,
- Desk-shape and Gutter-shape

Conventional gully tops are relatively inconvenient to handle and are therefore not particularly maintenance-friendly as far as the user is concerned. Handling is frequently rendered more difficult by fixed screw joints.



Installations





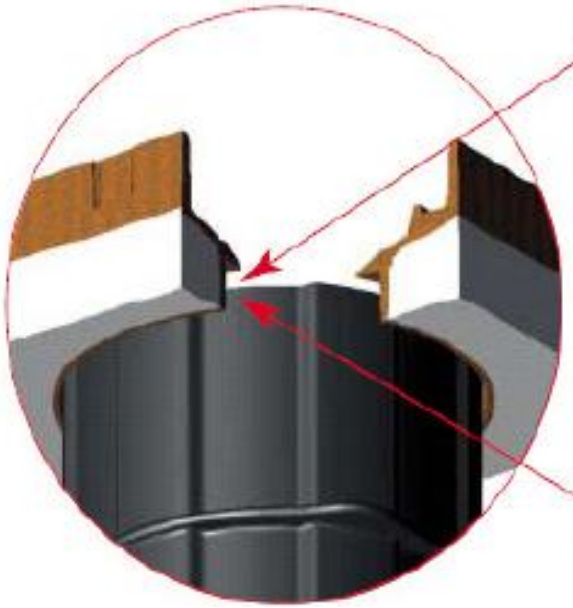
Combipoint

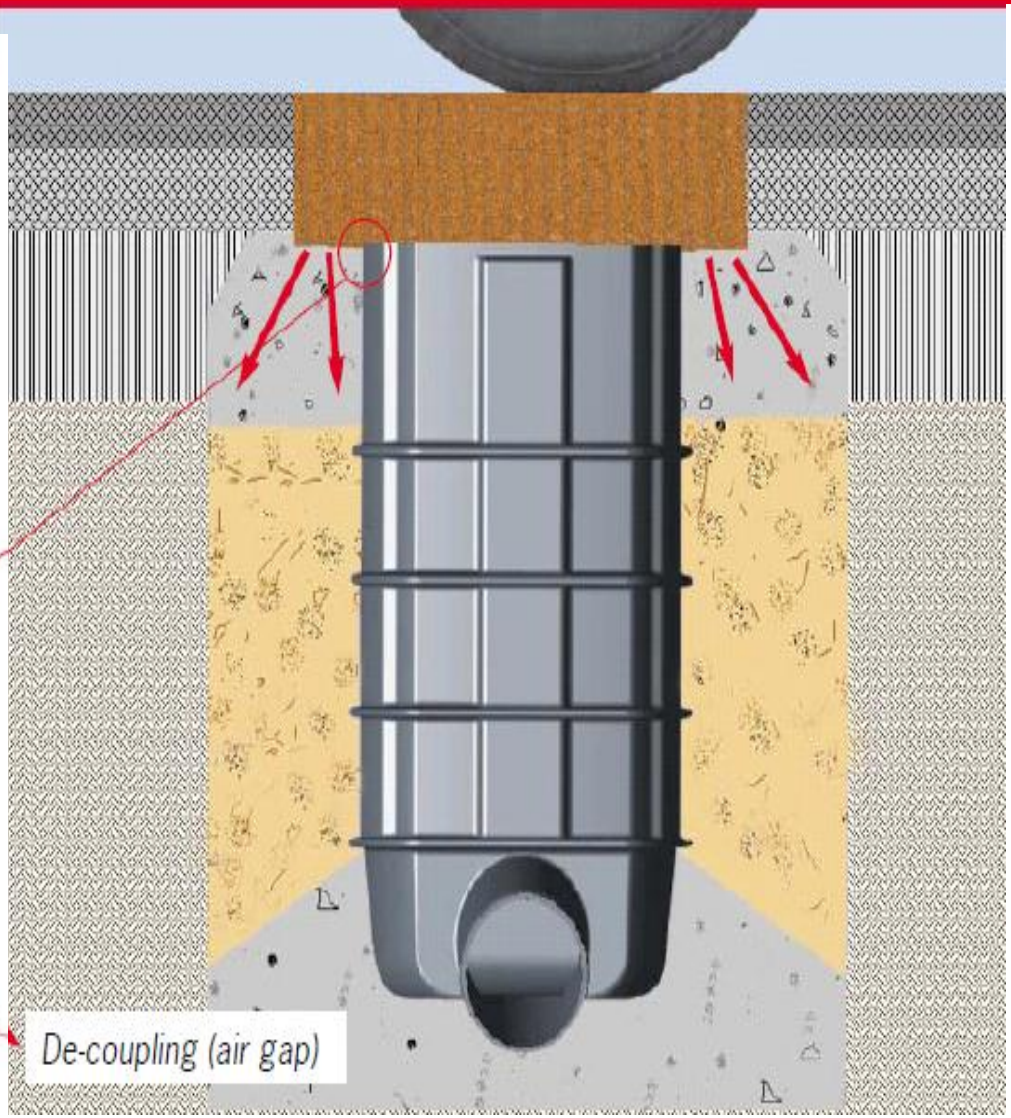
- No direct load transfer in the gully unit
- One-piece, permanently sealed gully unit no mortar joints
- Reduction in construction costs
- Simpler and easier installation
- No repair costs





- *Gully unit and grating frame de-coupled*
- *No mortar joint*
- *Avoids settling*
- *Water-tight gully*





Hydraulics



ACO Application team Provides

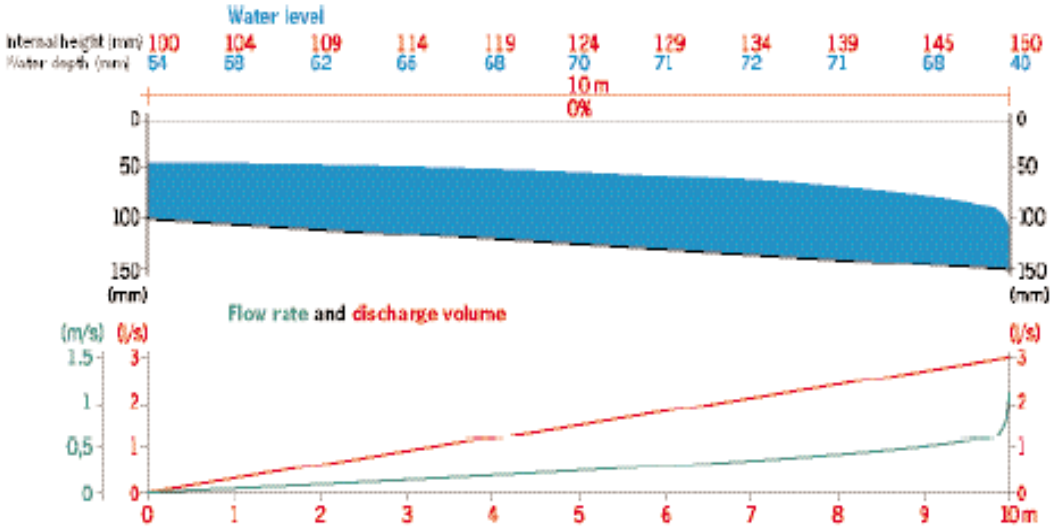
Precise hydraulic calculations,

tender specifications,

layout plans and installation supervision

$$Q = \frac{A \times r_{t(n)} \times \Psi}{10.000}$$

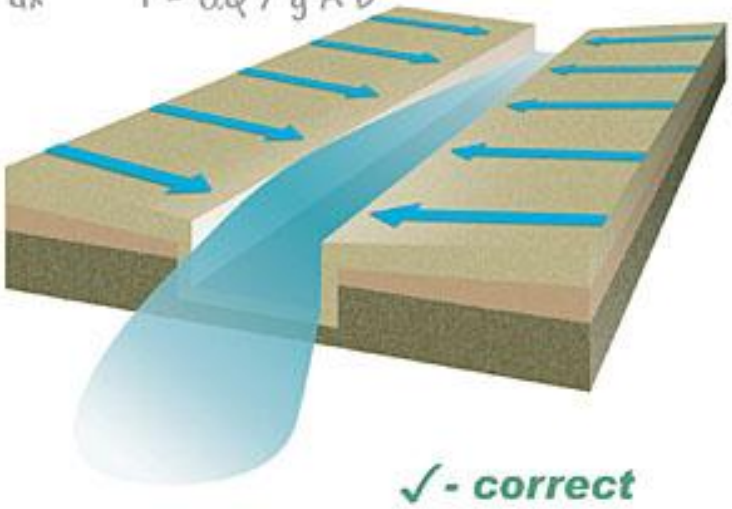
A = catchment area [m²]
$r_{t(n)}$ = rainfall [l/(s·ha)]
Ψ = drainage coefficient [-]



Hydraulic Performance of trench

NON UNIFORM FLOW

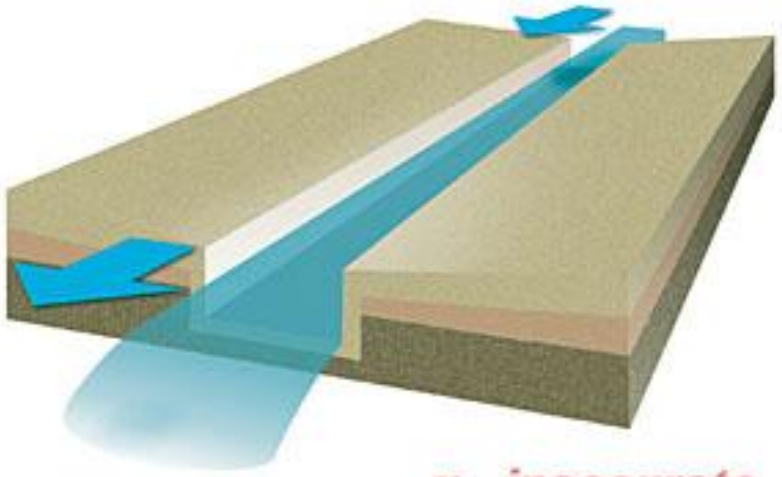
$$\frac{dy}{dx} = \frac{S_o - S - 2\alpha Qq / g A^2}{1 - \alpha Q / g A^2 D}$$



✓ - correct

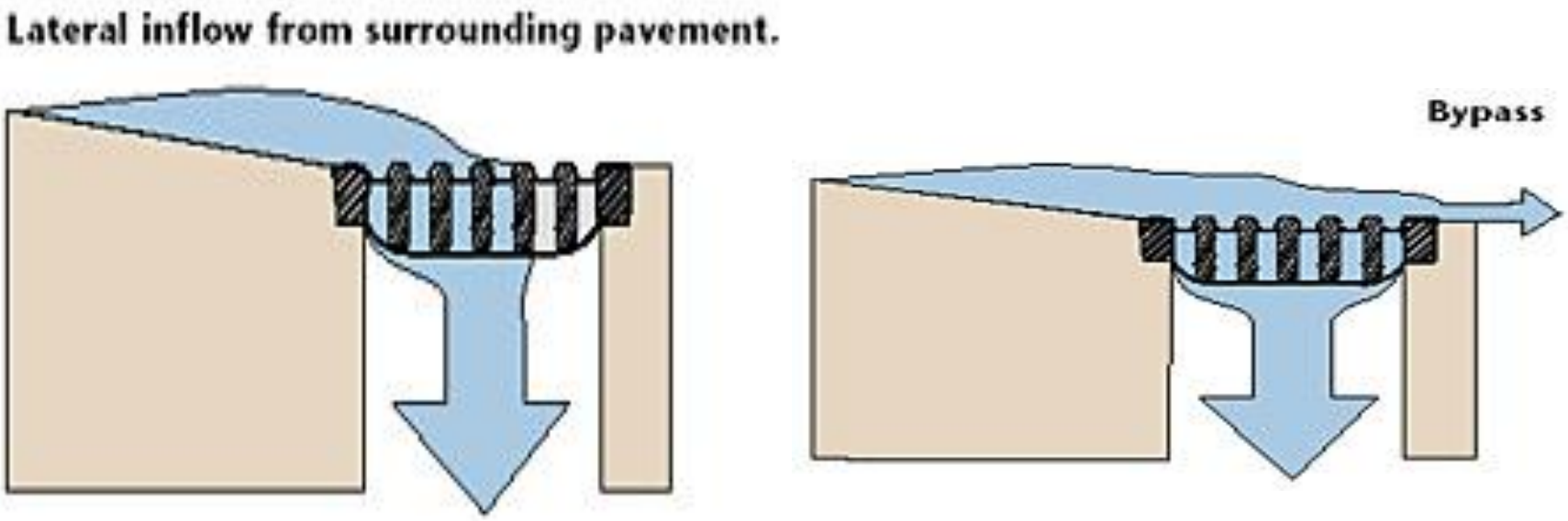
STEADY UNIFOARM FLOW

$$Q = A \times V \left(1/n \times m^{2/3} \times s^{1/2}\right)$$



x - inaccurate

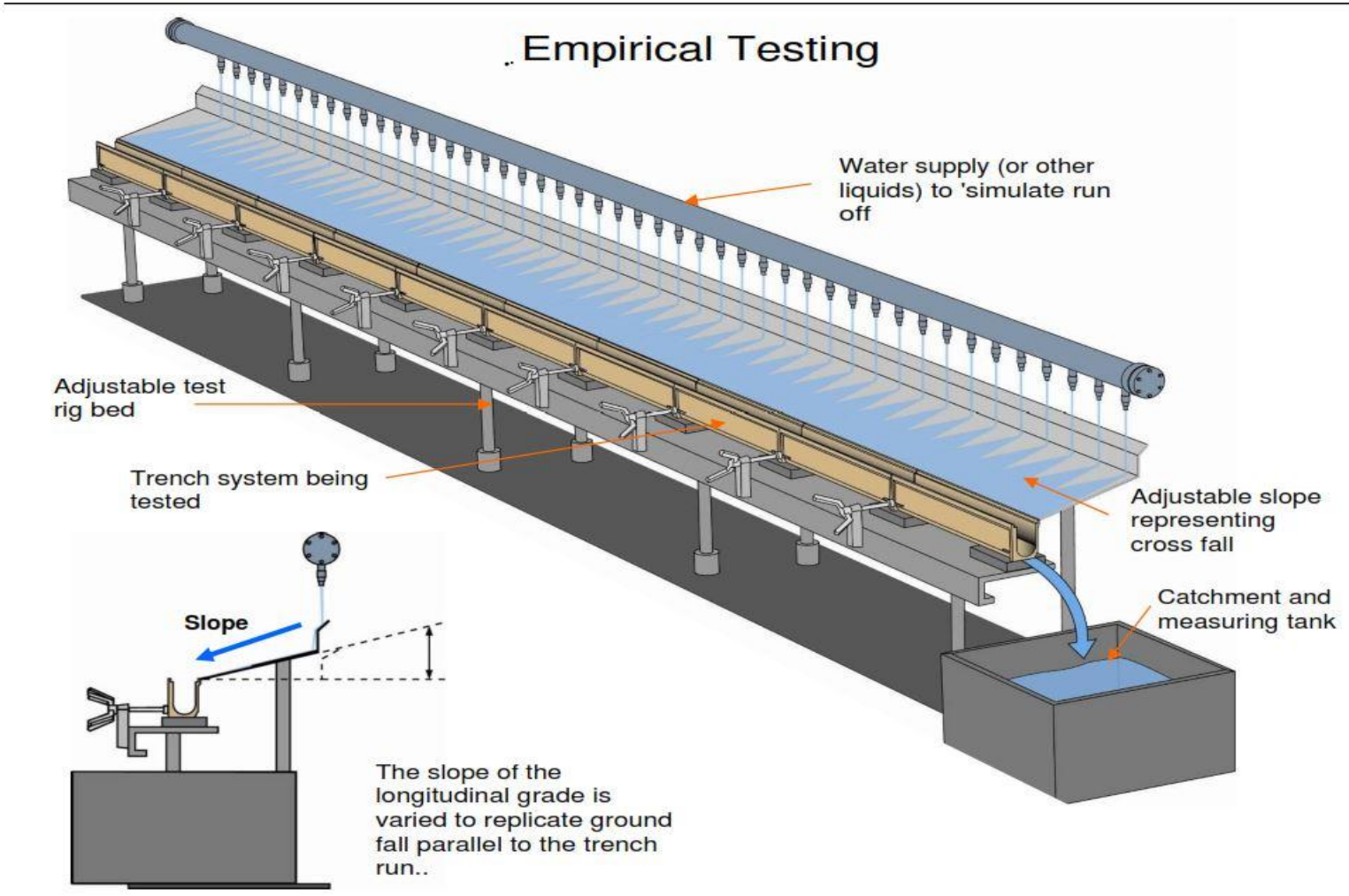
Grate Hydraulics



100 % CAPTURE

LESS THAN 100% Capture

“ACO- Hydro’’ Programe



“ACO- Hydro’’ Programme

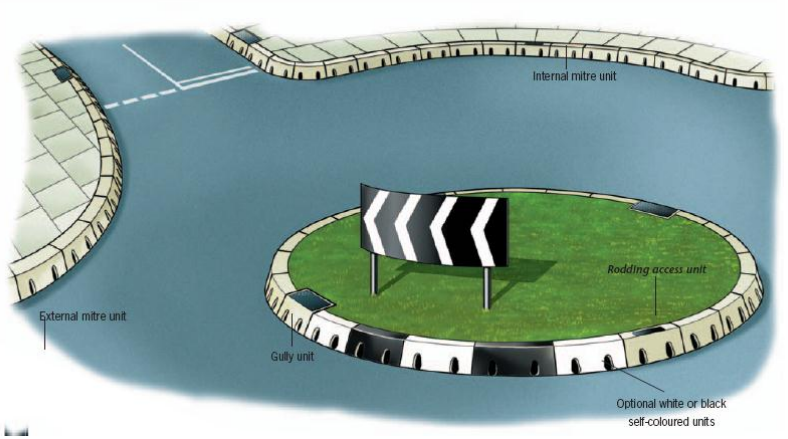


LANDSCAPE SOLUTIONS

Landscape - Kerb Drain

ACO KerbDrain™ is a one-piece resin concrete kerb and drainage unit for carriageways, roundabouts, local road schemes, car parking and many other applications.

ACO DRAIN® KerbDrain









Landscape - Kerb Drain

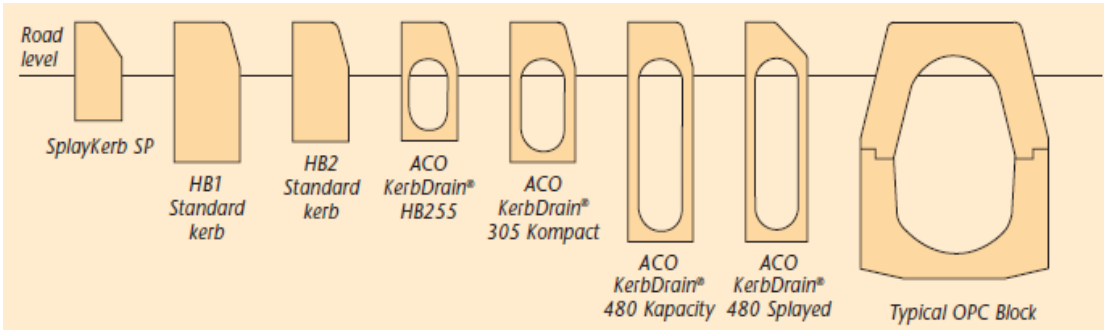
The system is suitable for new and renovation projects and has a full range of product accessories to provide a comprehensive drainage solution



Cable ducting

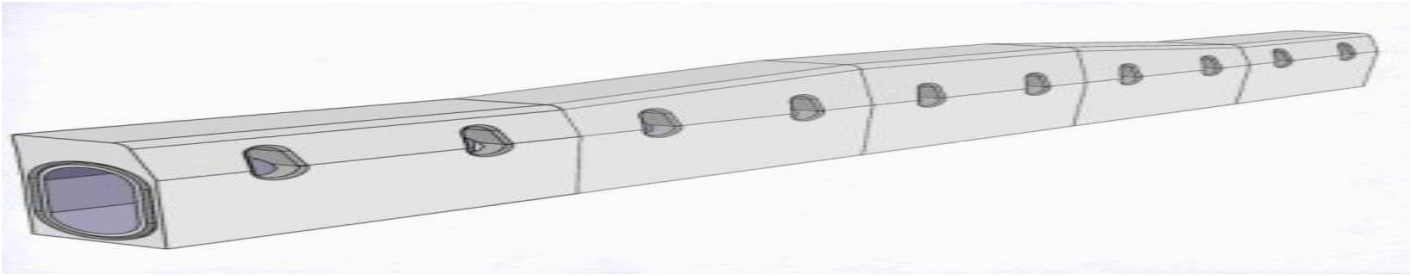
Blind kerb units can also be used separately as cable ducts for TV, cable, telecommunications etc





Landscape - Kerb Drain

Kerb Drains – Drop Stones



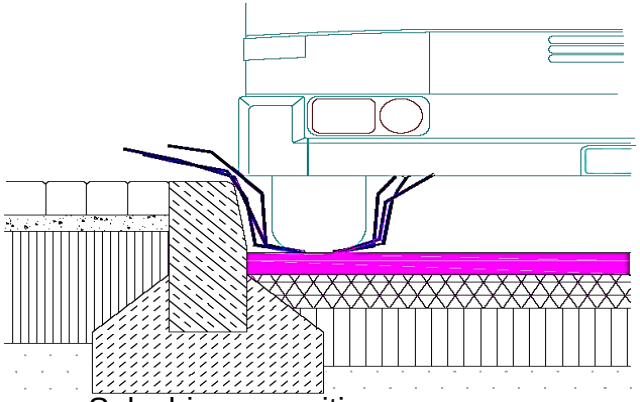




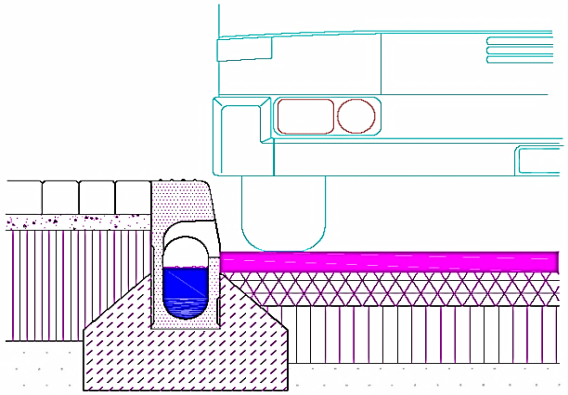
Landscape - Kerb Drain

Kerb Drains – For Bus Stops





Splashing on waiting passengers



No splashing on waiting passengers

Kerb Drain Installations – CWG Delhi



Kerb Drain Installations – CWG Delhi



Kerb Drain Installation- Amanora Pune



Kerb Drain Installation- Amanora Pune



Kerb Drain Installation- Amanora Pune



Kerb Drain – Knoor Bremse – Hinjewadi







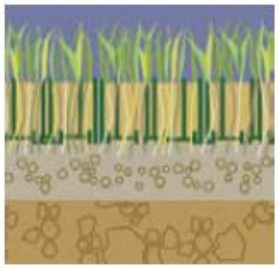
Kerb Drain Installations



Landscape - ACO Grass Grids

ACO Ground Guard grass reinforcement gives strength and good looks of a lawn.

ACO Ground Guard reduces the rate of surface water run-off, saving drainage installation costs.









Landscape - ACO Grass Grids



Landscape – Kalpataru Project



Landscape – Kalpataru Project



Landscape – Shantiniketan Project



Landscape - ACO Tree Guard

Tree Protection System







ACO Tree protection systems, comprising tree grilles and trunk protectors, ensure that the trees' roots will be provided with water and air and the trunks will be sheltered from traffic damage.

The root area is protected from soil compression, and the tree trunk from damage. The integrated watering cover facilitates watering and fertilizing, ensuring the survival of trees during dry spells.

Landscape - ACO Tree Guard

Tree Protection System

















Channel with Light Point

Architecturally attractive solutions ACO DRAIN® light points as combination of design and drainage in the entrance area of an office building





Application Areas

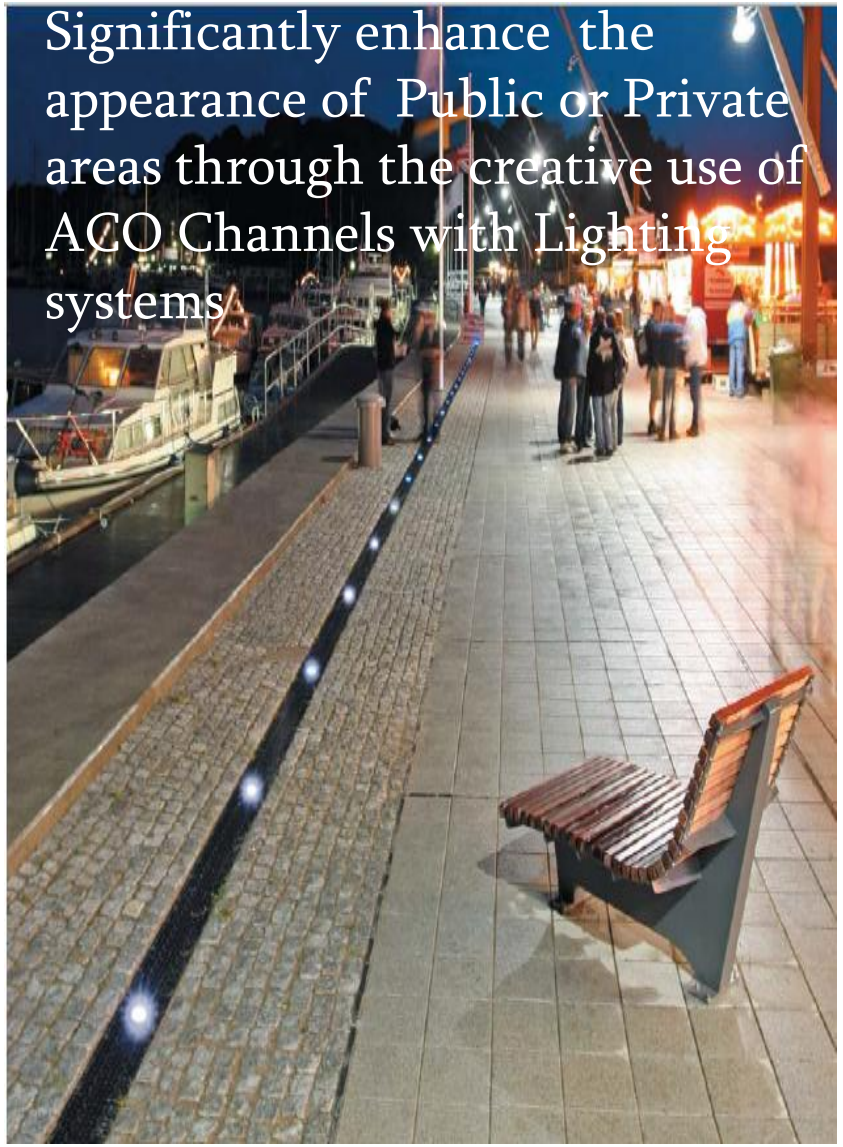
- Entrances,
- Promenades,
- Courtyards and
- Landscaped areas,



ACO Channel Lighting Systems



The systems combine the distinctive lighting feature with the efficient removal of surface water



ACO Channel Lighting Systems





Slot Drainage

Surface Water Drainage Systems – Drains with Brickslot

Medium Load Class – For Aesthetics, Safety, Discreetness, Pedestrian Areas















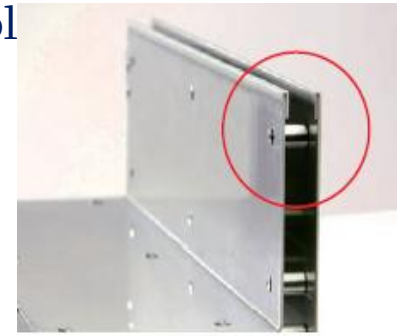
Slot Drainage

ACO Brick slot®

- Subtle and unobtrusive drainage system
- Offers high hydraulic efficiency in paved areas.

Ideal for shopping precincts, pedestrian areas, hotels, commercial and publ buildings.

The channel system comprises
slotted steel cover frame, with a locking device, fitted over a channel drain unit to form an integrated channel drainage system







ACCESS COVERS

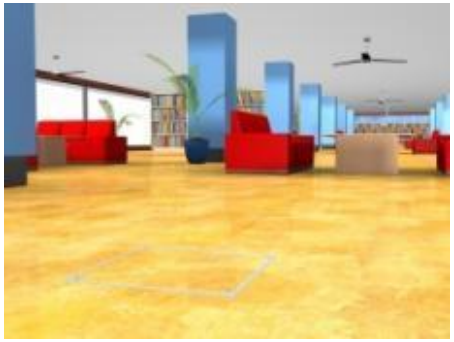


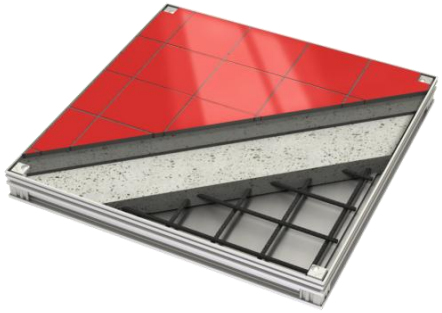


Access Covers

Access Covers – Light Duty

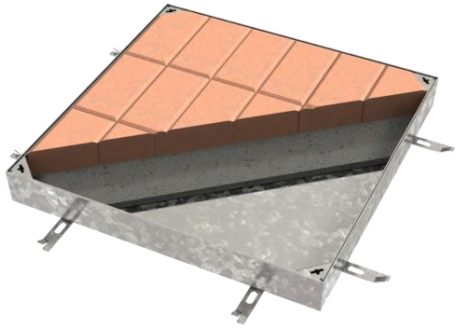




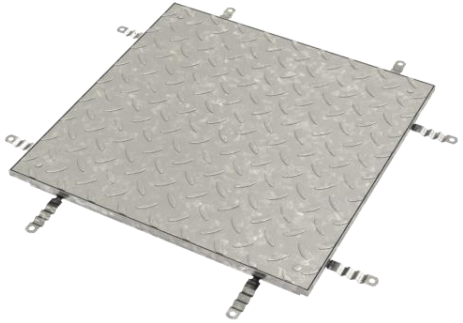
UNIFACE

- Recessed access cover for thin floor finishhing
- Must be filled-up with concrete in order to achieve requested loading class
- Selectable floor finishing in thickness up to 15 mm



PAVING

- Recessed access cover for pavement or natural stone infill in thickness up to 80 mm
- No concrete necessary

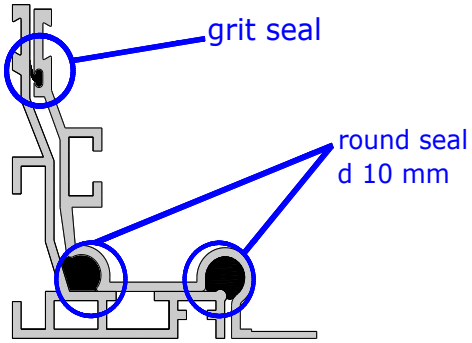


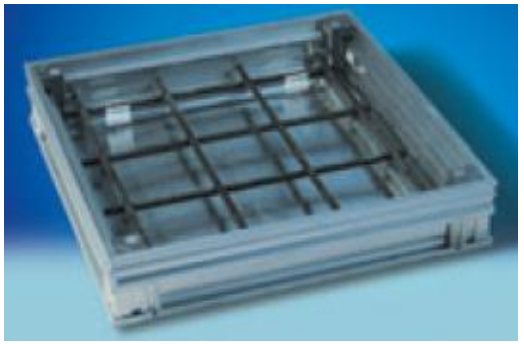
SOLID

- Solid top access cover
- Metal sheet with anti-skid surface on the top

Access Covers







Salient Features

- Made of SS 304,Galv. Steel, Aluminium
- Gasket sealed ,screw fixed & delivered with lifting handles
- Gas springs available for special application covers.
- Available in multipart for bigger dimensions
- Various sizes starting from 300x300mm to 1500 x 1500 mm & 1000 mm x 2000 mm
- Suitable upto D400 load class

Access Covers India Installations



P R Builders - Mumbai



Vijay Mallya Residence - Mumbai



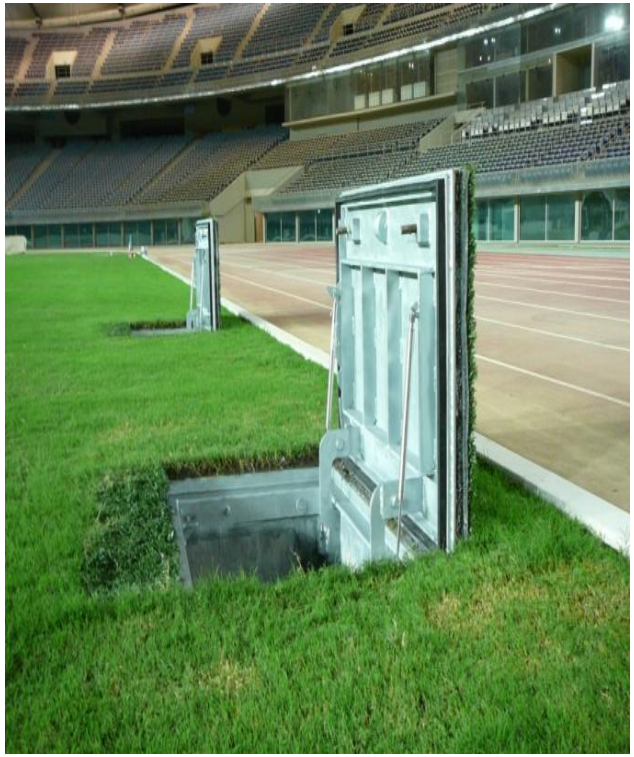
IGI Airport - Delhi

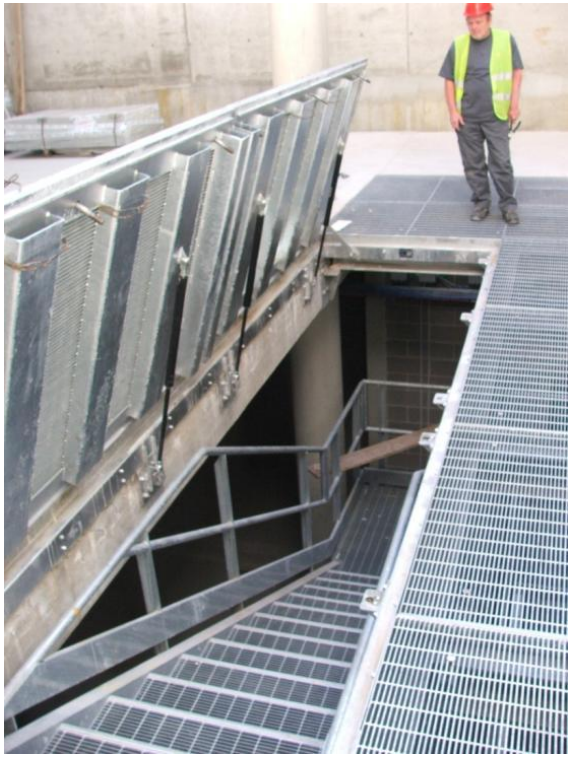
Access Covers India Installations



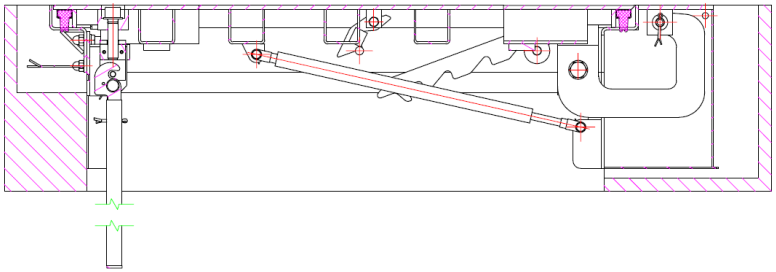


Access Covers – Heavy Duty







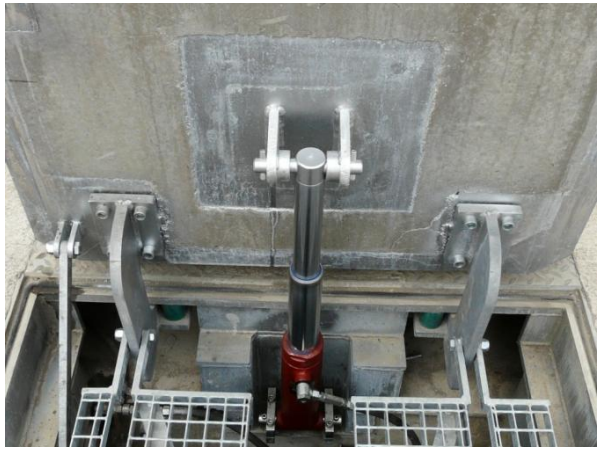




Access Covers – Heavy Duty

Access Covers – Heavy Duty







ACO Building Drainage -
Stainless Steel Drainage Systems











Application Areas

- Commercial Kitchens
- Canteens
- Food Preparation Areas
- Laundry
- Labs
- Hospitals
- Beverage Processing Area





- *'Kitchen Hygiene'-the **measures and conditions** necessary to control **hazards** and to ensure fitness for human consumption of foodstuff*

- *"Food business operators are to comply with appropriate measures to **control contamination** arising from the **air, soil, water**, and the storage, handling and **disposal of waste**"`*

FSA Guidelines

Rooms where food is prepared, treated or processed

Floors

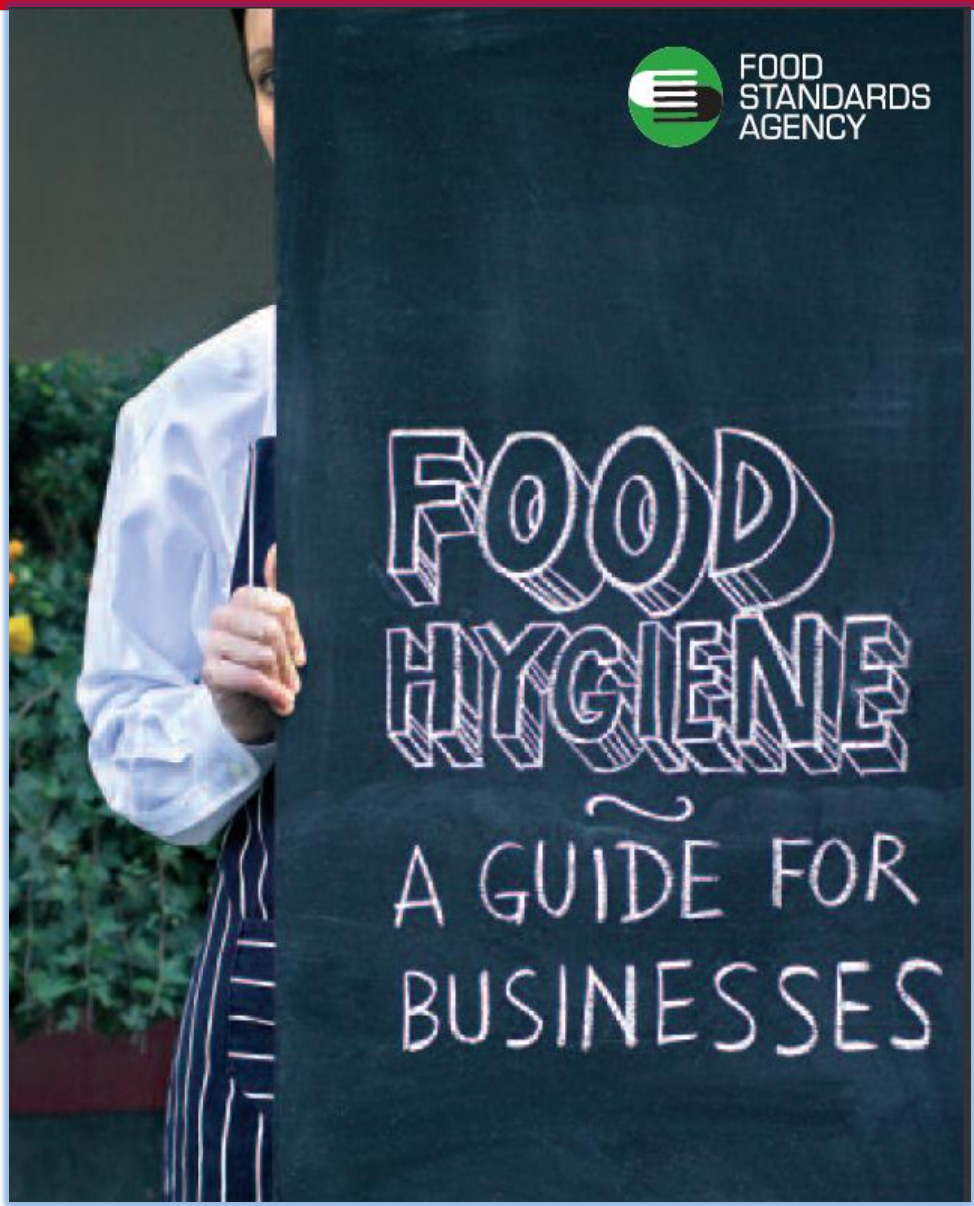
> Floor surfaces must be maintained in a sound condition and be easy to clean and, where necessary, to disinfect.

> This means that floor surfaces need to be made of materials that are impervious (i.e. do not allow fluid to pass through), non-absorbent, washable and non-toxic, unless you can satisfy your local authority that other materials are appropriate.

> Where appropriate, floors must allow adequate surface drainage.

Other requirements

> Drainage facilities must be adequate for the purpose intended. They must be designed and constructed to avoid the risk of contamination.



Adverse Effects of Poor Drainage

- **Bacterial contamination in Kitchen**
- Food is affected by pathogenic (disease causing) bacteria like
  - Listeria Mono cytogenes,
  - Salmonella and Fungi
- Which are most often detected in drains
- Grows in temperatures 0-45°C and attaches to stainless steel surfaces





Unhygienic Kitchen Drains…..

Key aim is to remove water from the cleaning process, especially in ‘high risk areas’!

- Improper Drainage leads to Growth of Microbial organisms and colonies **- Bio Films**

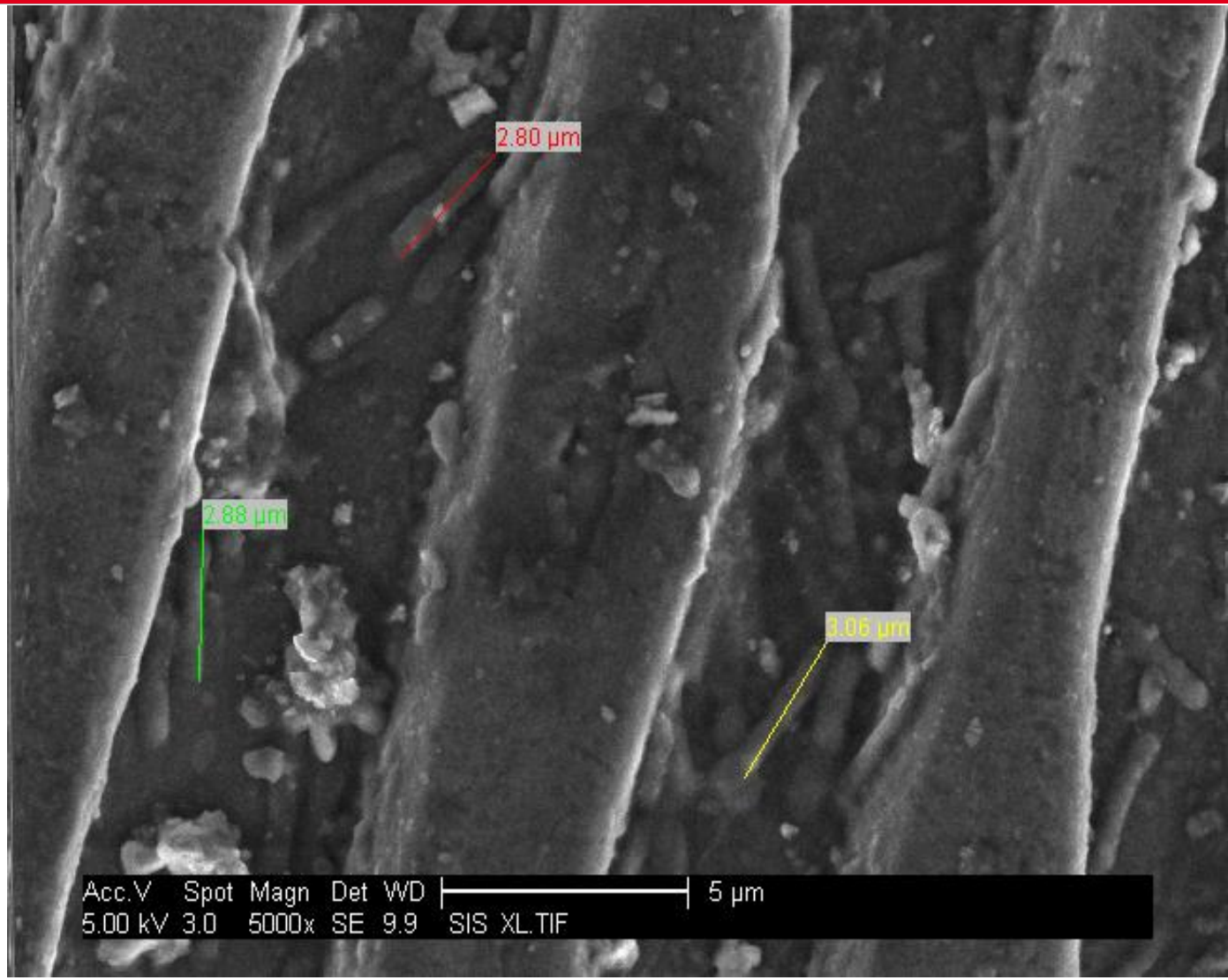




Electron microscope bacteria photographs

Bio-film development

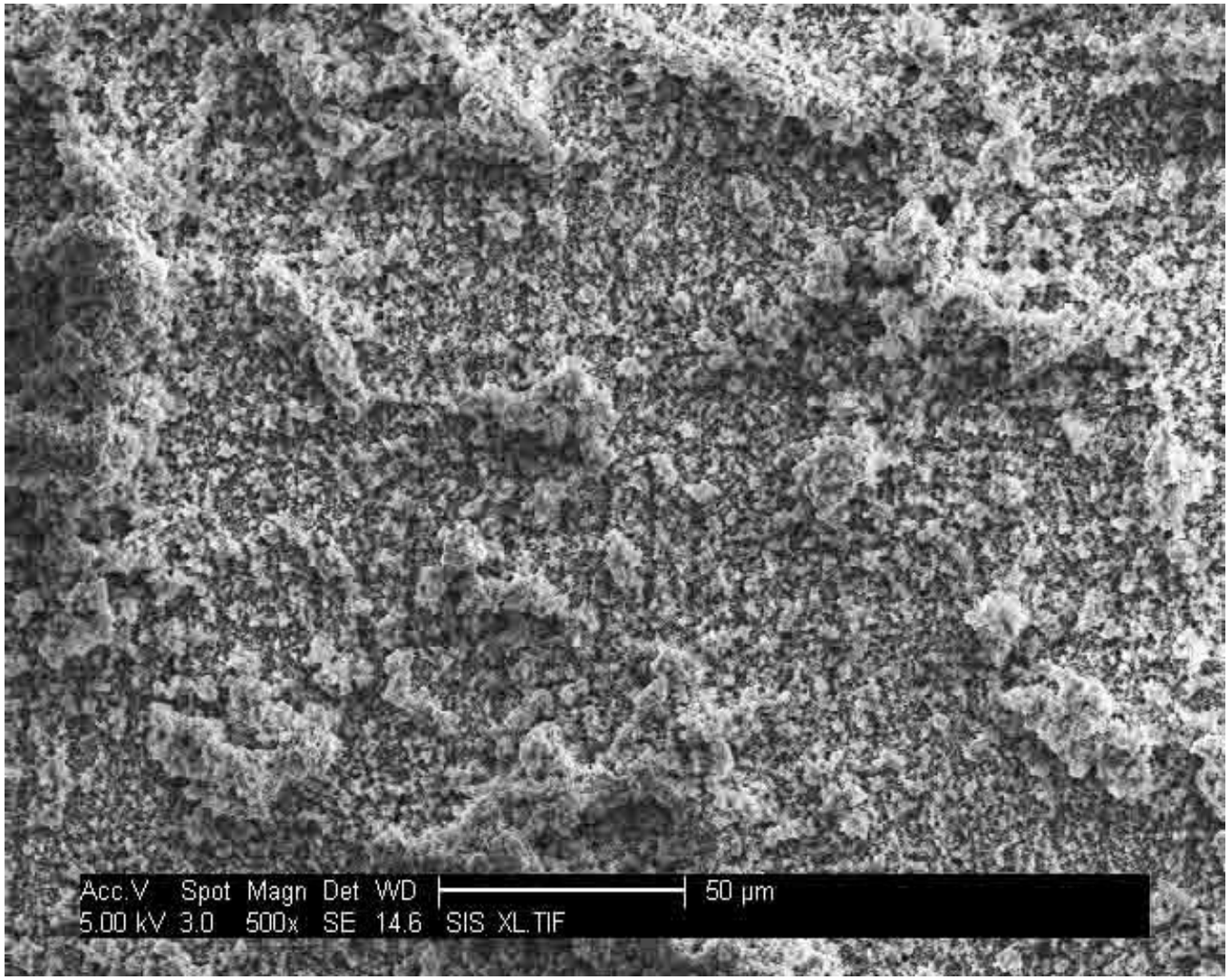
1st week



Biofilm development

Bio-film development

3rd week



Bacteria and biofilm development in un-hygienic kitchens



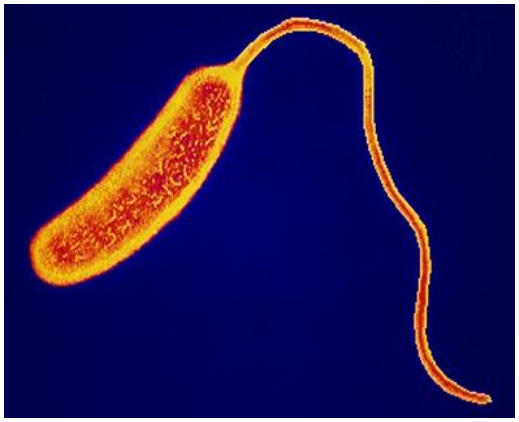


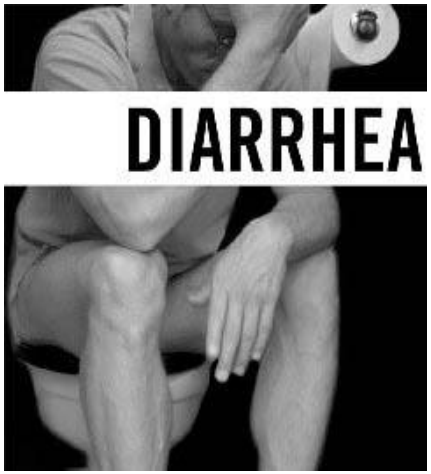


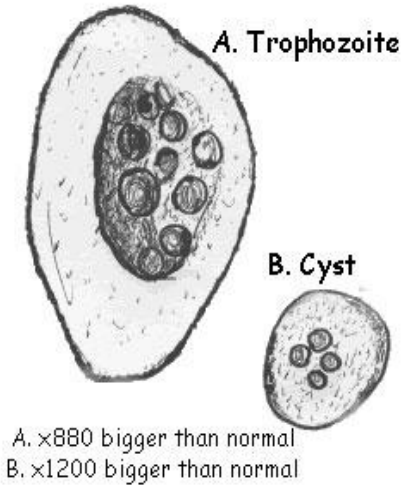


Consequential health hazards …….

- Amoebiasis
- Giardiasis
- Cholera
- Gastroenteritis
- Hepatitis
- Diarrhea
- Abdominal Cancer

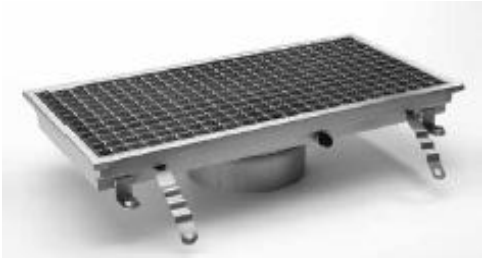




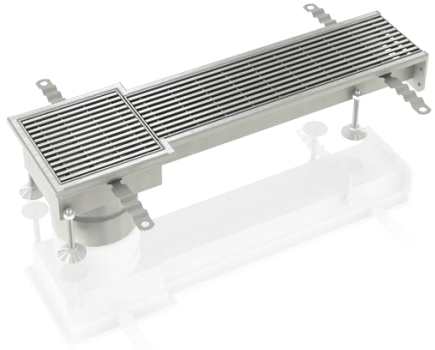


Possible Solutions……..

- ACO products are Designed & Engineered



- to ensure maximum hygiene as per international standards
- to **avoid contamination**
- facilitate **better flow rate**
- avoid **surface permeability** issues
- surface treatment of material for longer life and better hygiene
- avoid foul stench
- avoid clogging & flooding
- Ensure hygiene for indirect waste disposal
- avoid slippage and there by avoid accidents
- avoid pitting etc of surface areas
- … and addresses many more areas of concern





ACO Product Range

As the largest producer of stainless steel drainage systems, ACO has the knowledge, working experience and global expertise having worked with all major groups to offer optimal solutions for every application

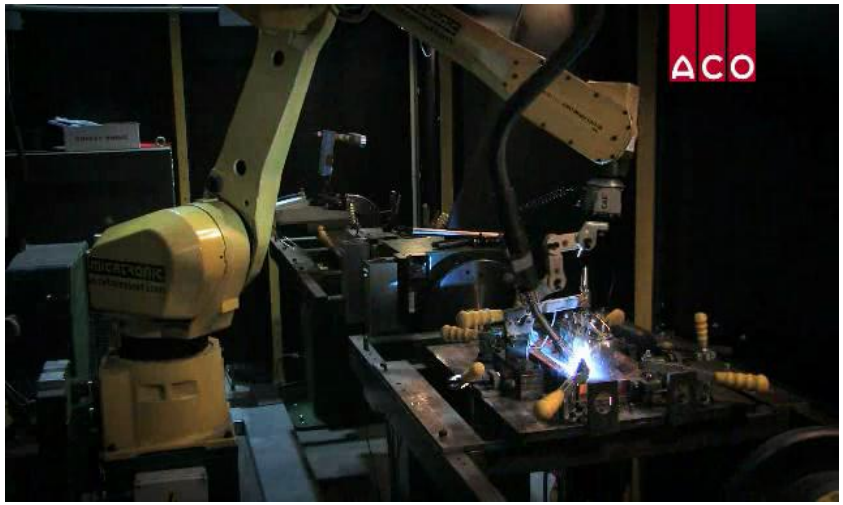


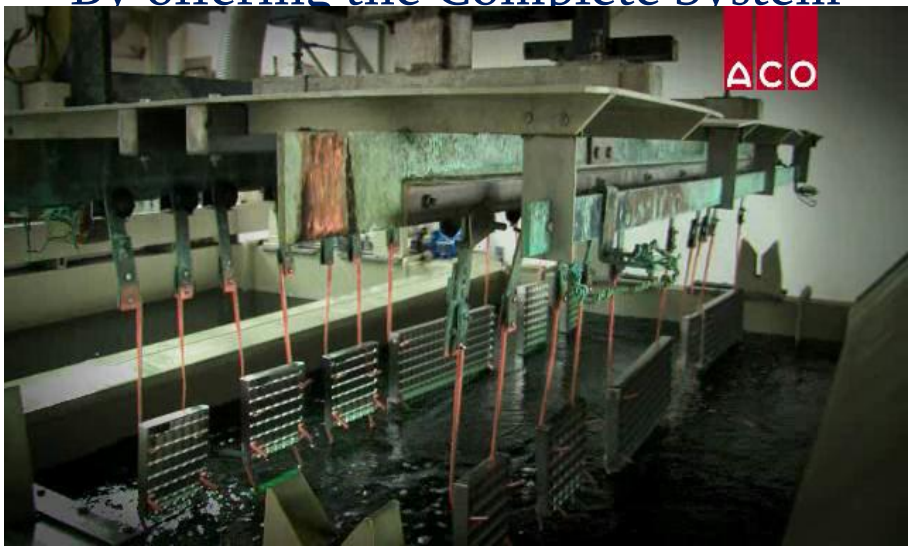
- Hotel Groups,
- Fast food chains,
- F&B industries, breweries& wineries,
- Dairies,
- Bottling plants,
- Food processing units,
- Marine applications etc…

ACO Stainless Steel Drainage Systems

Professional solutions from ACO…

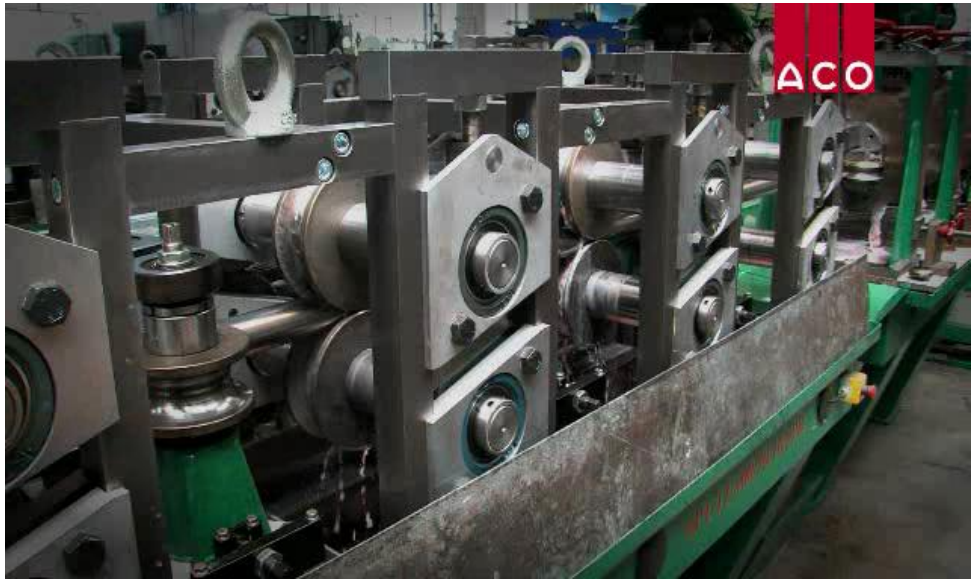
- By offering the best material SS 304 /SS 316
- Surface Treatment of finished product
- By offering the Complete System

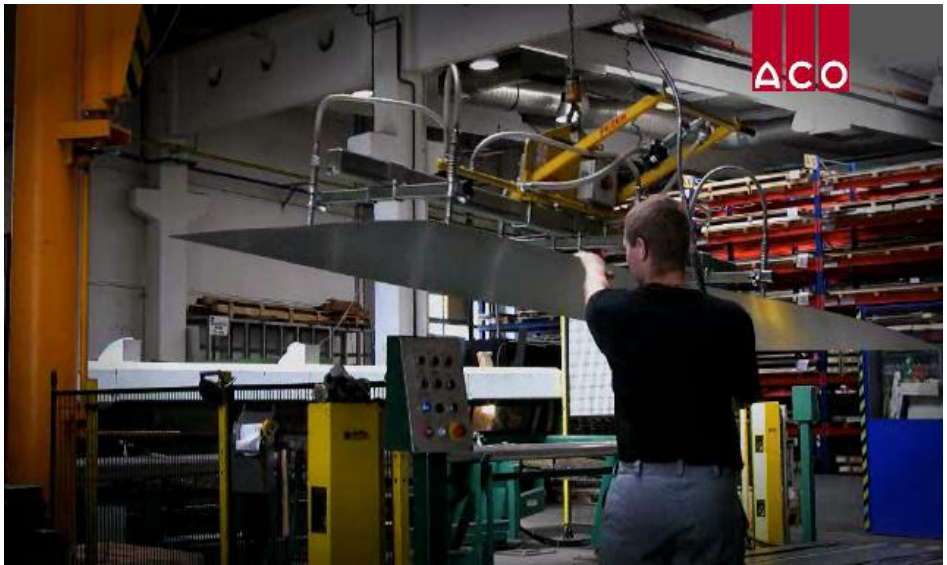






The Stainless Steel Material





Austenitic steels:

- Highly corrosion resistant: food processing, chemical/pharmaceutical & general industrial applications. *Non-magnetic*
  - Grade ‘304’: general purpose steel with excellent corrosion resistance
  - Grade ‘316’: superior corrosion resistance to Grade 304 for particularly aggressive applications

Surface Treatment

Surface treatment of stainless steel

The process of cutting, forming and welding of stainless steel will introduce impurities into he surface of the material and unless the appropriate action is taken, the material will begin to corrode and ultimately fail in service.



Pickling : Immersing finished product in a mixture of Nitric Acid and Hydrofluoric acid

Removes : Bluish high temperature oxide film and the chromium depleted layer

Passivation : Immersing finished product in Nitric Acid only

Removes : Embedded iron impurities and restores passive layer

Result :

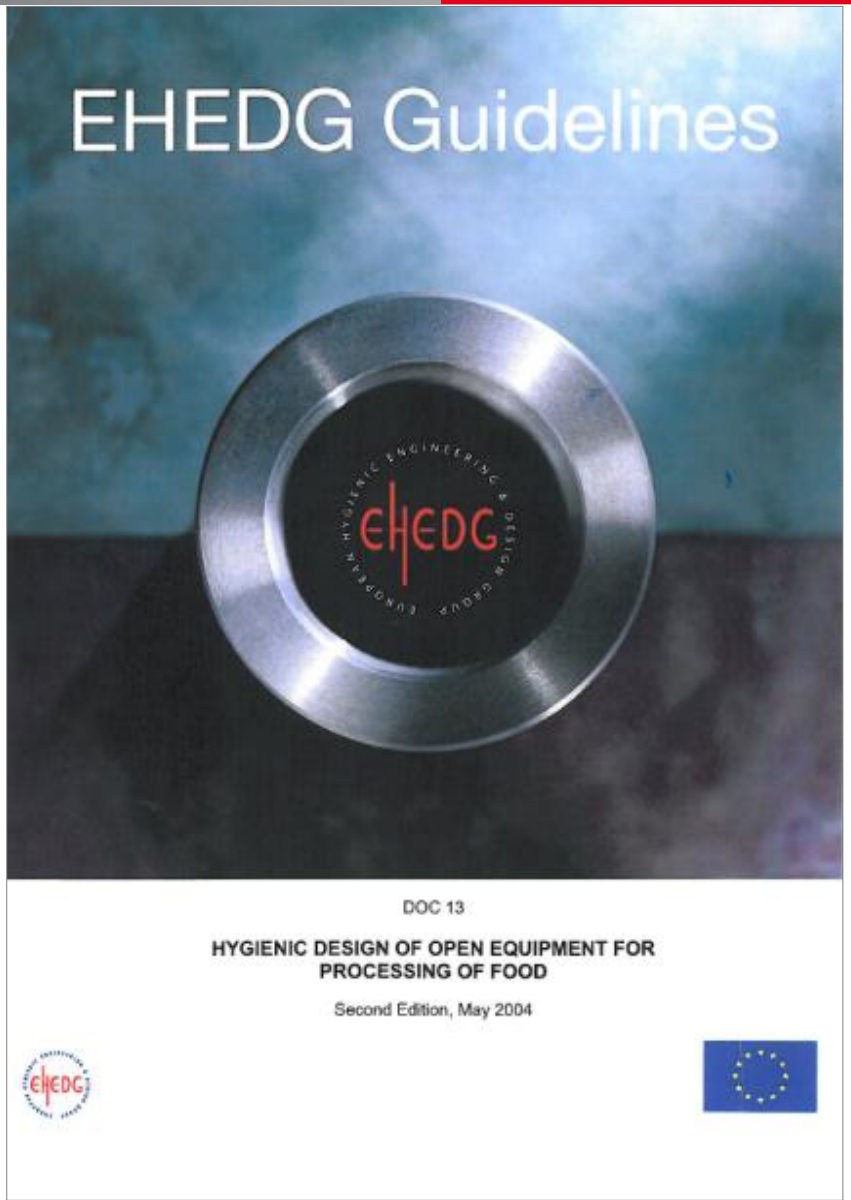
- Corrosion resistance of stainless steel is fully restored to its original state,

- Ensures long and reliable life performance together with aesthetic appearance.



Hygienic design

Food safety



- Applying standards reserved for food contact surfaces EN 1672, EN ISO 14159 and hygienic principles recommended by EHEDG to our product design.
- EHEDG Stands – Environmental Hygiene Engineering Design Group
- Safe connection with surrounding floor
- Advanced manufacturing technologies to ensure durability and special surface treatment to guarantee corrosion resistance
- Created together with key players in the food processing segment

Nestlé

Unilever

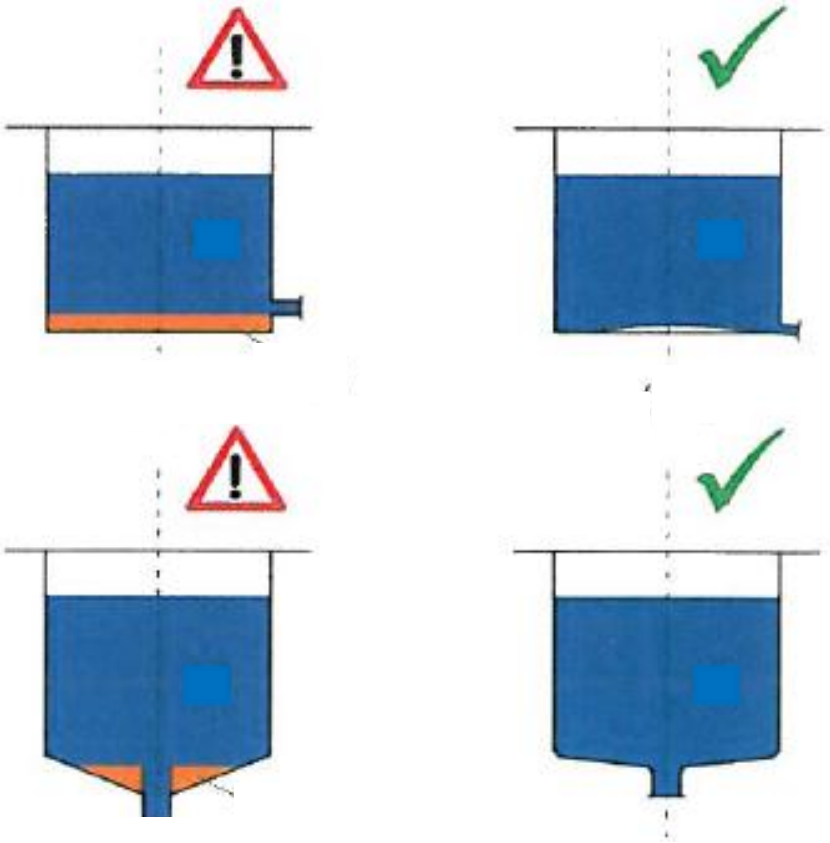
Hygienic design – basic principle

- Drainability
- Elimination of “metal to metal ” contact
- Rounded Internal corners

Hygienic design – basic principle

Drainability

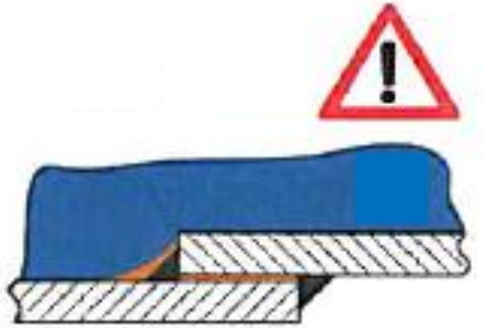
- Discharge outlet above the lowest level of equipment
  prevent self draining
- Self – drainable design with outlets at the lowest level +well rounded corners

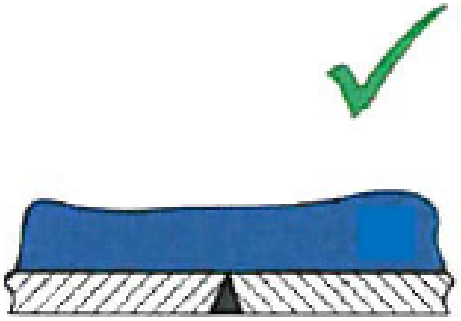


Hygienic design – basic principle

Elimination of “ metal to metal ”contact

- Permanent joints should be preferebly welded
- Overlapped joints and „skip“ welds must not be used

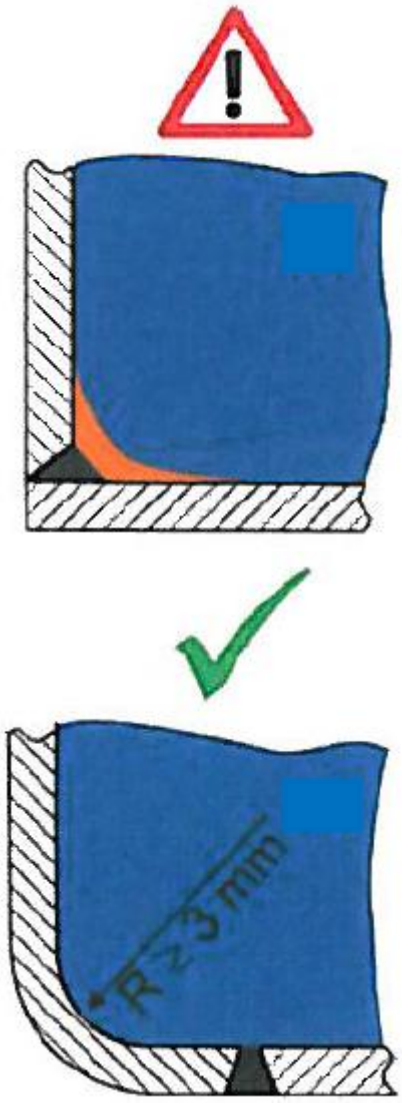




Hygienic design – basic principle

Rounded Internal corners

- Internal corners shall be designed using radii (Rmin = 3 mm)
- Welding in corners must be avoided
- Welding seems in the plane area are recommended



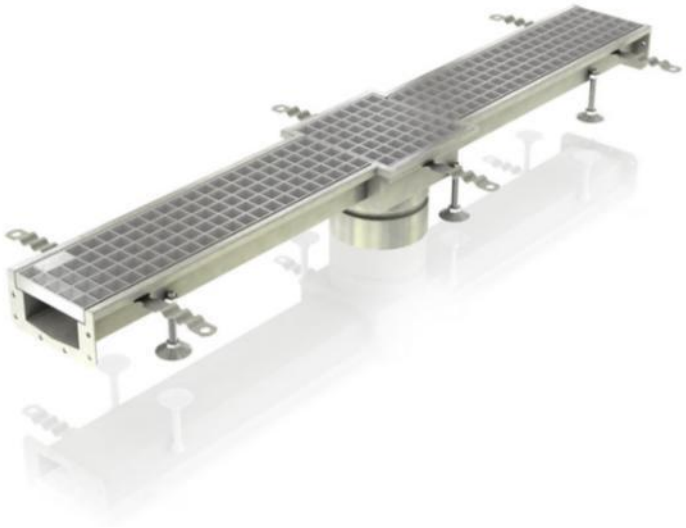
Hygienic design



ACO Stainless Steel Drainage Systems



Tray Channels - Box



Line drainage - Modular

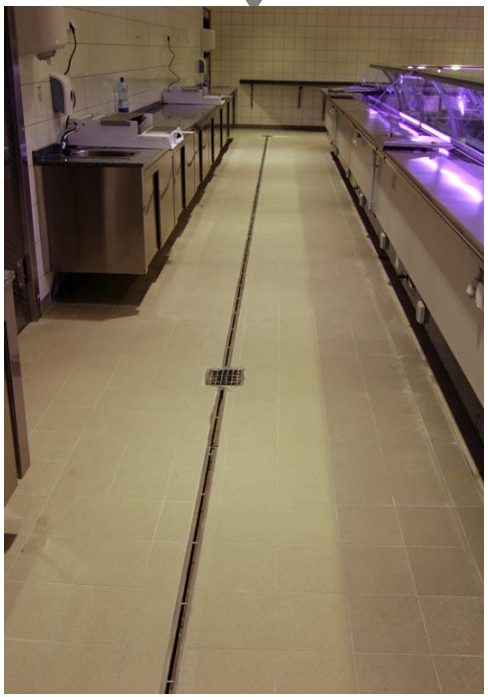


Point drainage - gully

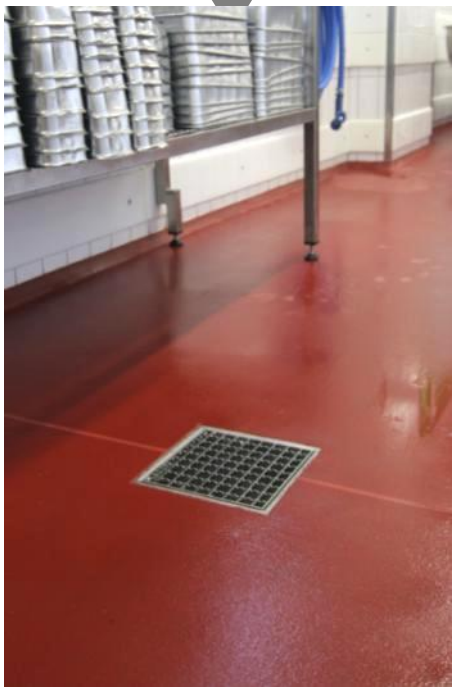
ACO Stainless Steel Drainage Systems – Standard Products



Tray Channels - Box

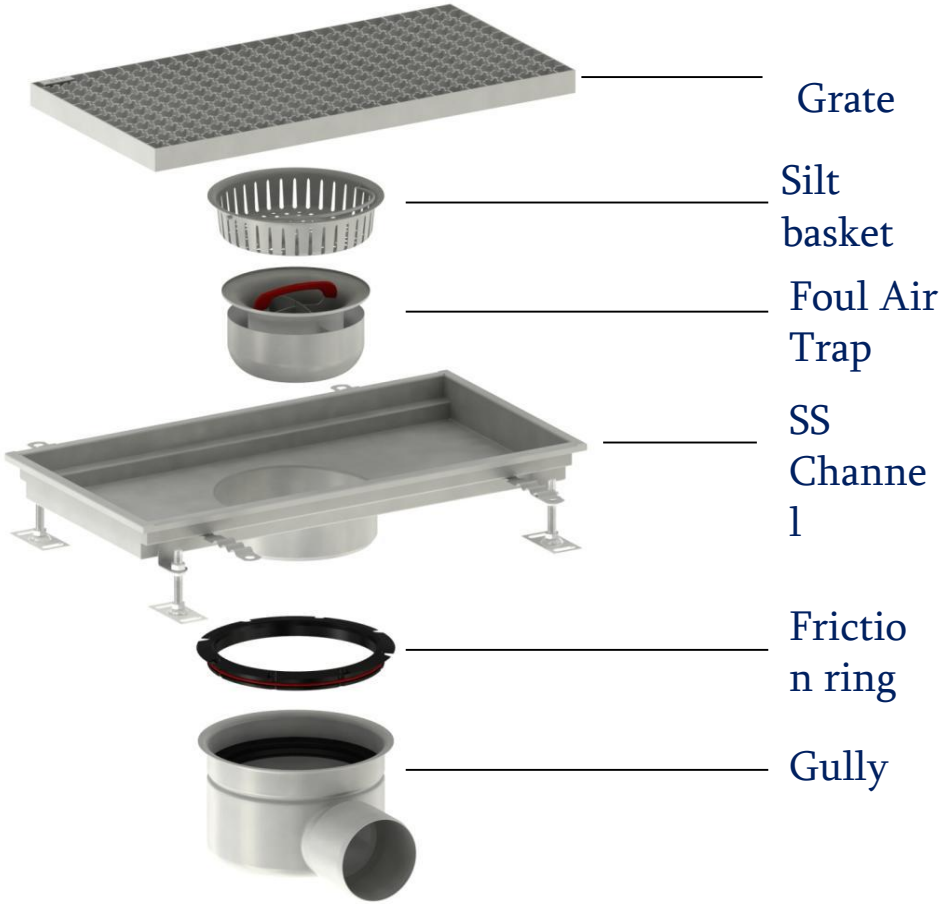


Line drainage - Modular



Point drainage - gully

System Overview and Standard dimensions



Load classes - standards

Application	EN 1433 (Drainage channels) & EN 124 (Manhole & Gully Tops)	EN 1253 (Gullies for Buildings)
	-	H1.5
	A15	K3
	-	L15
	-	
	B125	M125
	C250	
	D400	
	E600	

Grates - Load class L15 EN 1253

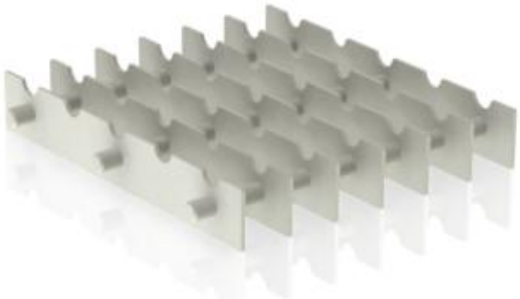
- Areas with light vehicular traffic in commecially used premises



Mesh grating



Heelsafe grating



Arla grating



Volcano grating



Quadrato

Grates - Load class M125 EN 1253

- Areas with vehicular traffic, such as car parks, factories and workshops



Ladder grating



Slot cover

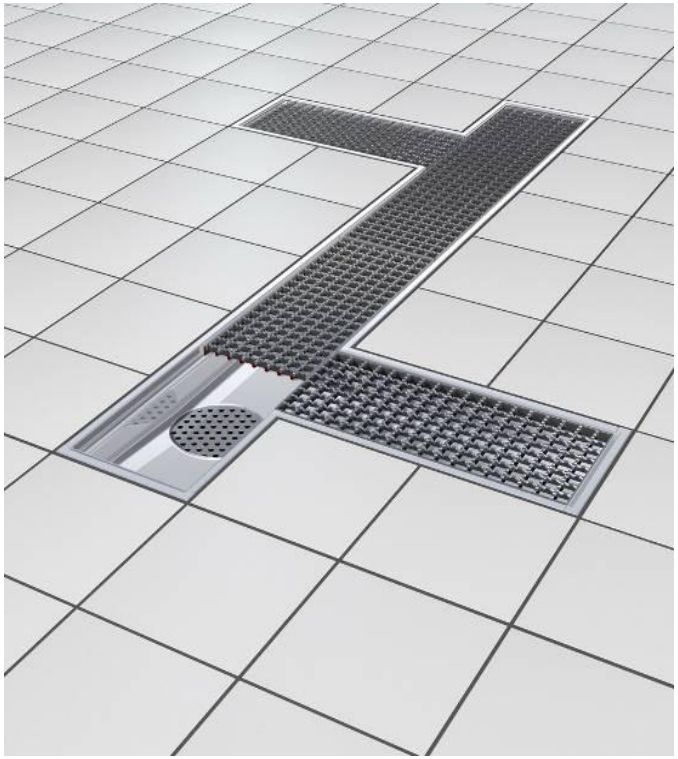
Grates - Load class C250 EN 124

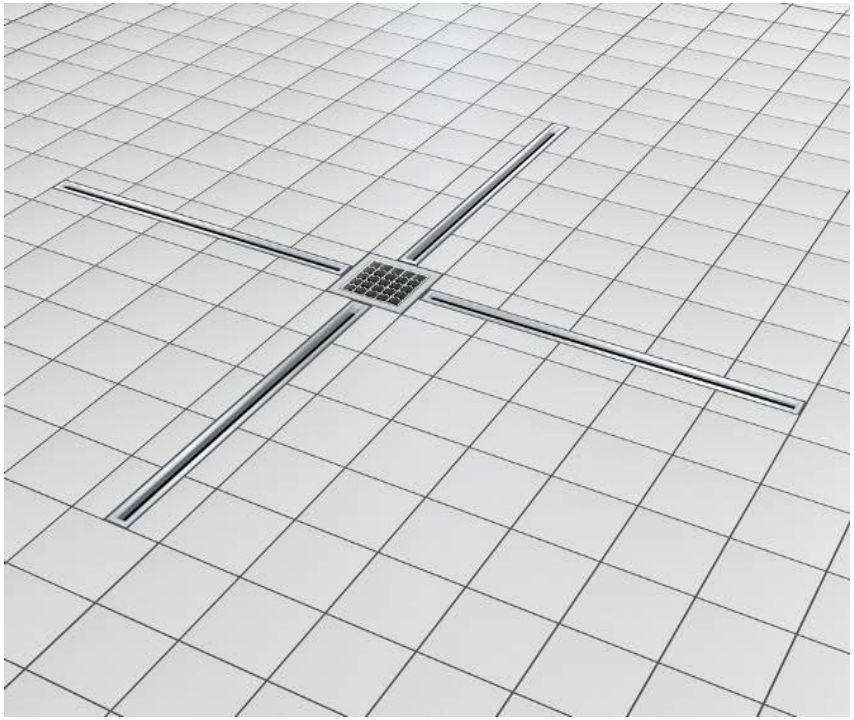
- Factories , industrial plants ,lorries, pallet trucks and small fork lift trucks



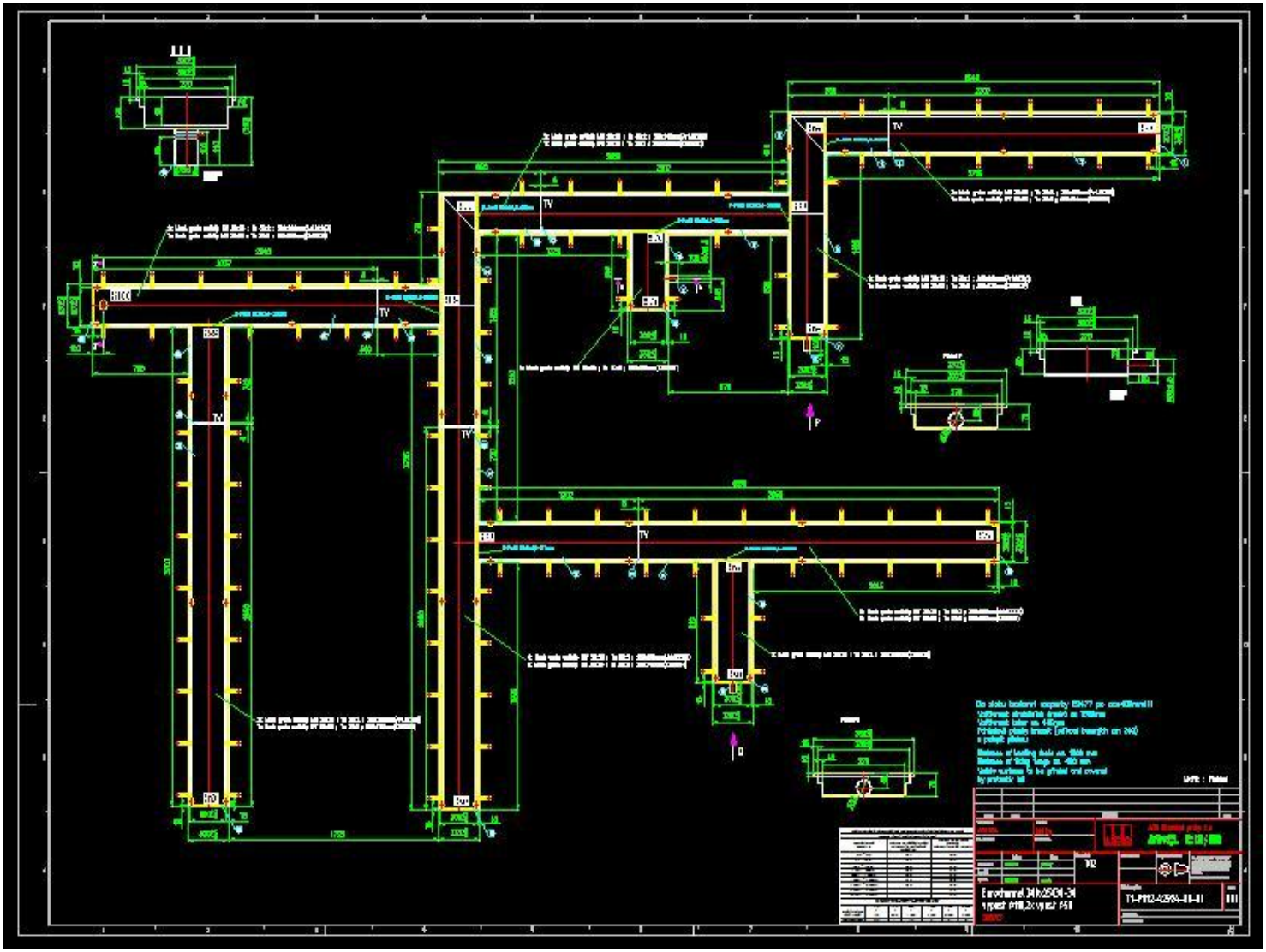
Ladder grating

…. Customised Solutions





….and Customization



ACO Kitchen Channel – Photographs





ACO Kitchen Channel – Photographs





ACO Hygiene Gully – Photographs





ACO Installations in India

Hilton Hotels, Chennai





Crowne Plaza Hotel, Delhi





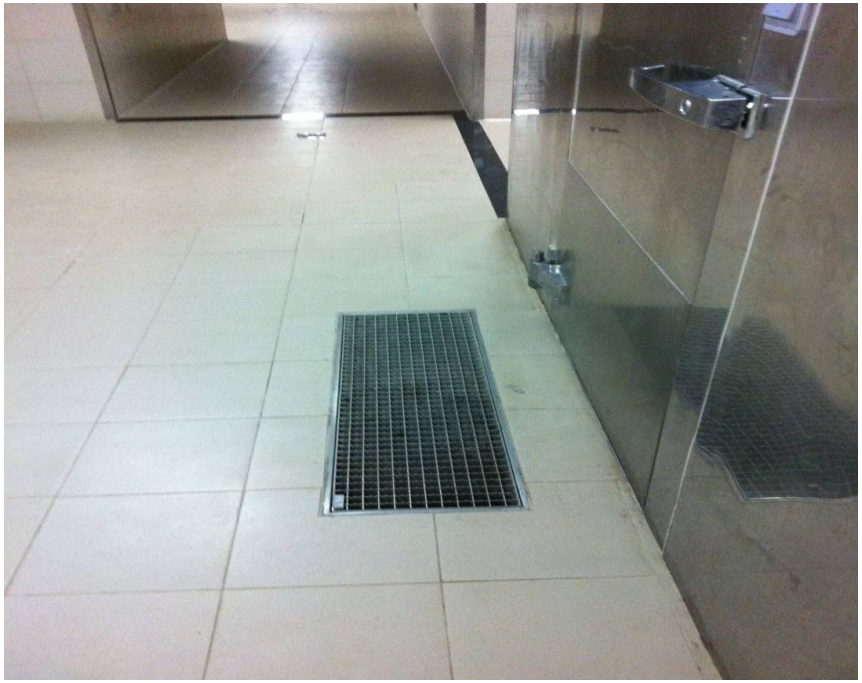
ACO Installations in India

Marriott Hotel, Jaipur





Movenpick Hotel,Bangalore





Stainless Steel Pipe Systems



Stainless Steel Pipe Systems





These features include:

- The low "first cost" choice of material for the expected life of the building.
- Cost competitive alternative to traditional cast iron and aluminium alloy systems.
- Excellent corrosion resistance provides for long-term contract dependability.
- Cost savings on fewer assembled parts.
- Unique hygienically designed couplings to reduce wastage during installation.
- Non combustibility of stainless steel pipe controls spread of fire risk through walls and compartments.
- Non-hazardous material.
- Absolutely weather proof and available electropolished finish to special order.

Stainless Steel Pipe Systems



ACO PIPE® stainless steel pipework systems are manufactured from austenitic stainless steel in grades 304 and 316. All products are chemically pickled and passivated for optimum durability and corrosion resistance.



ACO PIPE® stainless steel pipes and fittings are available in 50 mm, 75 mm, 110 mm, 125 mm, 160 mm and 200 mm external diameters with pipes supplied in standard from 0.15 metre up to 6 metre lengths for optimum practicality and ease of assembly.

SEPARATORS

ACO GREASE SEPARATORS

Waste Water Containing Grease Appears In…

- Hotels
- Restaurants
- Industrial Canteens
- Food Courts
- Staff Canteens/Kitchens









ACO GREASE SEPARATORS

Waste Water Containing Grease leads to



…Pipe Obstruction.



…Formation Of Corrosion And Noxious Stench.



…Negative Effect On Treatment Plants.



Conventional Grease Separators









- Un Hygienic
- Grease separation not adequate
- Clogging of pipes
- Effect on downstream treatment plants
- Very strong and pungent odour
- Failure of STP and WTP

ACO Grease Separators

- ACO is one of the largest manufacturers of Grease Separators in the World
- manufactured according to the applicable standards of EN 1825 and DIN 4040-100.
- hydraulically tested and have a general official approval from DIBt, Berlin
- inspection by the Landesgewerbeanstalt Bayern, Germany to check separator production for adherence to the currently applicable test standards.
- LGA certified
- GET certified
- KIWA certified

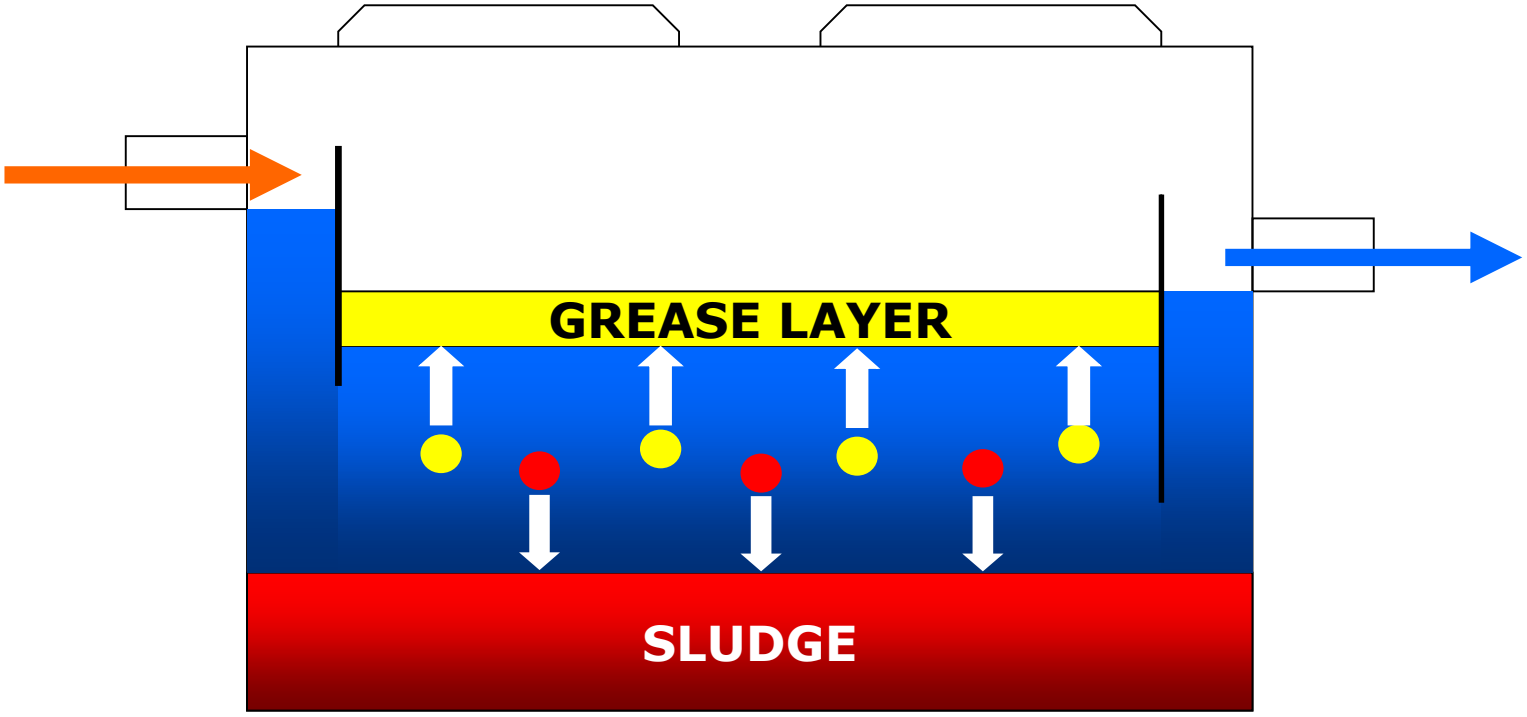
ACO Passavant separator plants are tested and monitored

LGA Landesgewerbeanstalt Bayern

checks separator plants for adherence to pertinent test standard.

Basics

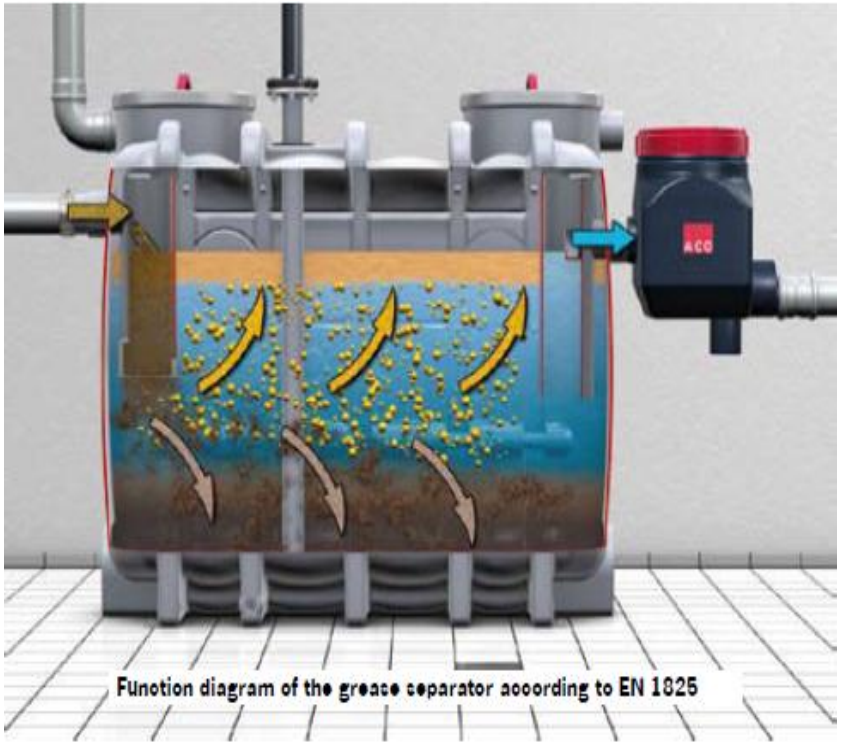
Grease separator - functional principle



Both activities happen synchronised. Clear water will flow out of the separator.

Basics

- Grease separator plants have to be used if animal and organic fats and oils have to be separated
- By means of gravity, grease separators shall prevent animal and organic fats from entering drainage objects.
- This applies to commercial and industrial establishments, where animal and organic fats occur.

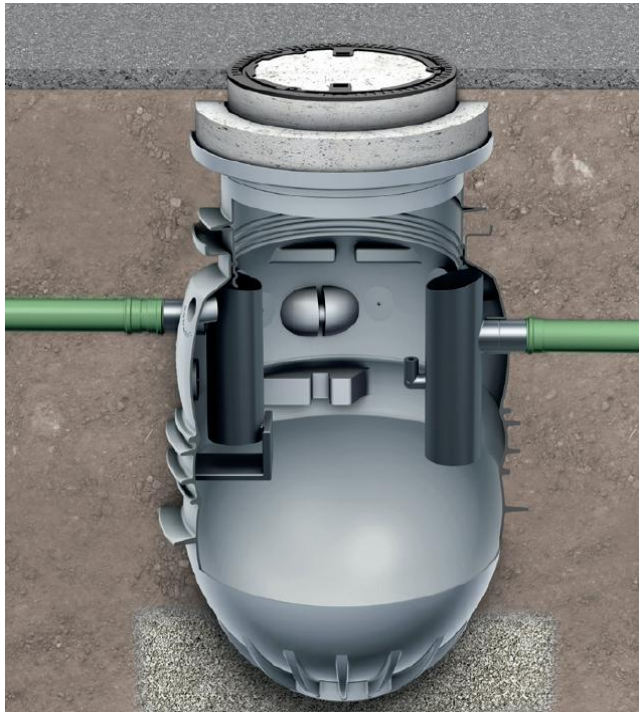


Function diagram of the grease separator according to EN 1825

Product range

Installation Options

Under Ground



Above Ground



Product range

Type Of Disposal

Full disposal	Partial disposal





Grease Separators – Eco Mobil 0.3 & 0.5

Grease Separators





ACO Separators - Under Sink Photograph



ACO Separators - Under Sink Installation Photograph

Grease Separators



ACO Separators - Under Sink Installation Photograph

Grease Separators





Separators



Separators

Grease Separators



Separators

Grease Separators



Separators



Separators

Grease Separators



Separators



Separators



Separators



Separators

Hilton Hotels,Chennai



Separators

IBM , Bangalore



Nokia,Chennai



Separators

Movenpick Hotel,Bangalore



Hozman Resorts,Coorg



Separators

Leela Hotels, Gurgaon



Shell Services, Chennai



Grease Separators – Below Ground Installation



Grease Separators – Below Ground Installation



Grease Separators – Below Ground Installation



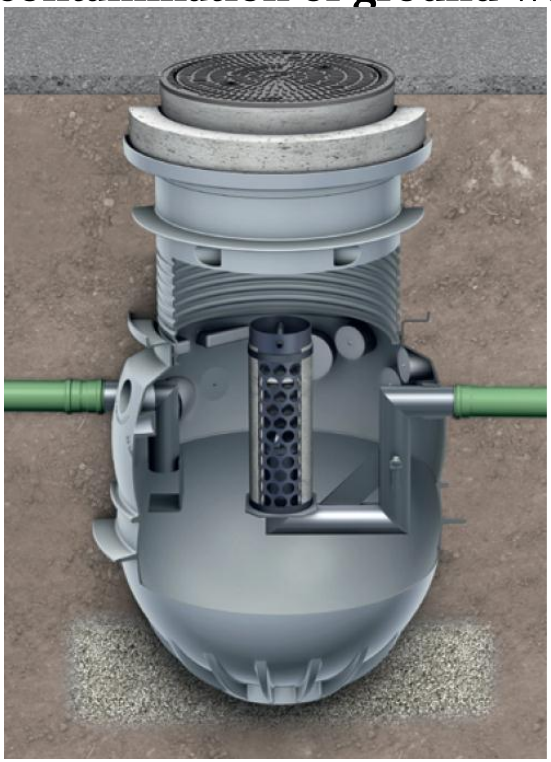
Grease Separators – Below Ground Installation



Oil Separators

Oil Separators

To prevent challenges with light oil such as inflammable and explosive atmospheres and contamination of ground water table



Application Areas

- Petrol Stations
- Parking Spaces
- Parking Areas
- Carriageways
- Car Wash Plants & Cleaning Sites
- Workshops
- Vehicle Recycling
- Heliports
- Transformer Substations
- Elevator Shafts





Class	Maximum permissible content of residual oil mg/l	Typical separating technique (for example)
I	5	Coalescing separator
II	100	Gravity separator

Oil Separators Inner Part – Siemens Power Plant Bharuch



Oil Separators Inner Part – Siemens Power Plant Bharuch



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Oil Separators Inner Part – Siemens Power Plant Bharuch







Oil Separators Inner Part – Siemens Power Plant Bharuch



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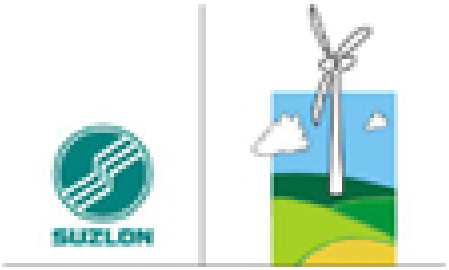
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Thank You